When Do Gender Quotas Change Policy?

Evidence from Household Toilet Provision in India*

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Abstract

While gender quotas in politics have been adopted worldwide, evidence on their impact on policies is fairly mixed. This could simply be due to differences in contexts or policies across studies. We use household level data on allocation of toilets for the entire rural population of Uttar Pradesh, the largest state in India (25 million households), and show that gender quota effect exhibits substantial heterogeneity within the same political context and for a politically salient publicly provided good that women prefer more. We exploit the larger gender gap in preference for toilets among Muslims than Hindus, due to cultural differences across religions, to show that variation in women's intensity of preference (relative to men's) drives the heterogeneity. Further, we find that women with stronger preferences are more likely to demand their preferred policy under female leadership, which explains our results. These results provide suggestive evidence for a less explored channel wherein greater expression of preferences by women under female leadership can shape the gender quota effect.

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1 Introduction

Quotas for women in political positions are a widely used policy aimed to improve their representation in politics. Currently, 130 countries across all the continents and covering both democracies and non-democracies have some form of quota for women in politics.[1] An extensive literature spanning across countries documents whether gender quotas in politics, or women in political positions more generally, matter for policy (Brulé & Toth, 2022; Cassan & Vandewalle, 2021; Brulé, 2020; Clayton & Zetterberg, 2018; Gajwani & Zhang, 2014; Clots-Figueras, 2012a, 2011; Ban & Rao, 2008; Chattopadhyay & Duflo, 2004). These studies are primarily motivated by the observation that preferences for public goods are gendered, i.e., women and men prefer different public goods (Sapiro, 1981). Hence, gender quotas should change policy, as elected women would prioritize the allocation of a different set of public goods than their male counterparts.

However, the evidence on the effects of gender quotas on policy is mixed. While Franceschet & Piscopo (2008) (Argentina), Braga & Scervini (2017) (Italy), Brulé (2020); Chattopadhyay & Duflo (2004) (India), Clayton & Zetterberg (2018) and Bhalotra *et al.* (2020) (cross-country evidence) find that women leaders pursue different policies, Van der Windt *et al.* (2018) (Congo), Devlin & Elgie (2008) (Rwanda), Bagues & Campa (2021) (Spain), Ferreira & Gyourko (2014) (USA), Bardhan *et al.* (2010) (India) find null effects. Given that the evidence comes from independent studies conducted in different countries with focus on different policies, the observed heterogeneity in the effects of gender quotas may reflect either different political contexts in which the quotas were implemented or variation in the political salience of the policies that the studies focused on. This makes it hard to draw general conclusions about the overall effectiveness of gender quotas in shaping women's substantive representation.

In this paper, we show that gender quotas in elections have substantial heterogeneous effects on policy even within a given political context and for the same publicly provided

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(private) good, allocated under a common program. We use data on local elections in Uttar Pradesh, the largest state in India (comparable to the fifth most populous country in the world) and allocation of a politically salient good for which women are known to have a stronger preference—namely household toilets. We show that variation in women's intensity of preference (relative to men's) for this good, due to differences in cultural norms across salient social groups, drives the heterogeneity. Finally, we show suggestive evidence that the "demand mechanism", whereby women with stronger preferences express higher demand for this good under a female leader, is a potential channel explaining our results.

Women's relative intensity of preference can be an important determinant of policy prioritization under female leadership. Clayton & Zetterberg (2018), for example, find that gender quotas in national legislatures across democracies increased public spending on health—an issue that women associate with more than men (Clayton *et al.*, 2019; Bhalotra & Clots-Figueras, 2014; Miller, 2008), but had no effect on education, for which gender gap in preference is found to be smaller (Gottlieb *et al.*, 2018; Clots-Figueras, 2012b; Chattopadhyay & Duflo, 2004). However, estimating heterogeneous treatment effects by comparing across public goods can be confounded by other factors. For instance, health policies that involve supplying supplemental nutrition or increasing local care workers are easier to implement compared to education policies, which might require more systemic or institutional changes. Our analysis, by focusing on a single publicly provided good, precludes this problem.

Additionally, various cultural considerations may result in differences in women's intensity of policy preferences, even on an issue traditionally associated with them. Greenberg (2001) argues that, in the context of the US, even though women on average are more liberal than men, religiosity especially among White women makes them more conservative, resulting in divergence of opinion across women. Junn & Masuoka (2020) point out that preference for the Democratic (i.e., the center-left) party in the US is significantly stronger among African-American women, relative to White women.

In a similar vein, cultural factors shape the gender gap in preference for toilets in rural

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India, even though women overall have a stronger preference for toilets than men. Using household level survey data on usage of and preference for toilets in rural North India (including the state of our study), we find that compared to men, women on average are 9 percentage points more likely to use toilets conditional on owning one and are 5 percentage points more likely to report toilets as their top ranked good that they wished to spend money on. However, Hindus associate in-home toilets with religious considerations of impurity and pollution, making them reluctant to either construct or use toilets within home premises (Coffey *et al.*, 2014, 2017). This makes Hindu men and women more aligned in their preference for toilets. Such concerns are, however, absent among Muslims. Consequently, the gender gap in preference for toilets is higher among Muslims than Hindus. We find that conditional on owning a toilet, Hindu women are 9 percentage points more likely to use it compared to Hindu men. For Muslims, this gap increases to over 14 percentage points, i.e., the gender gap in the usage of toilets is 62% higher among Muslims than Hindus.[2] At an aggregate level, villages with a higher population share of Muslims have a greater overall gender gap in preference for toilets. We, therefore, use the population share of Muslims in a village as a measure of relative intensity of preference for toilets among women and test whether the effect of gender quotas in toilet provision is different in villages with a higher Muslim share. We discuss below that other village characteristics that might be correlated with Muslim population share do not explain our results.

We estimate the effect of female quotas in the 2015 village council president (or, *Sarpanch*) elections in the state of Uttar Pradesh (UP), India on the subsequent provision of household toilets in 2016–17. The toilets were allocated under the *Swachh Bharat Mission* (SBM) or 'Clean India Mission', launched by The Government of India in 2014. The village governments (Gram Panchayats or GPs) were the primary implementing agencies of this program. We use household toilets as our target good due to a combination of factors. First, it is well known that women have a greater preference for toilets than men (Coffey *et al.*, 2014; Khanna & Das, 2016; Stopnitzky, 2017). Moreover, a lack of sanitation facilities has been associated

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with the widespread practice of open defecation in rural India. Access to safe sanitation through reduction in open defecation has been shown to have significant positive effect on development outcomes such as child health and women's safety (Guiteras *et al.*, 2015; Kumar & Vollmer, 2013; Hossain *et al.*, 2022). Consequently, the construction of toilets has recently received a lot of emphasis from policymakers, making it a politically salient publicly provided good.

We compile household-level data on allocation of toilets for the entire rural population of UP (over 25 million households) and match it to election results for the universe of about $59,000$ GPs in the state. We identify the religion of the household heads from their names using a religion prediction algorithm (with more than 97% accuracy) developed by Chaturvedi & Chaturvedi (2023). The female quota in UP is not randomly assigned, which makes the sample of GPs with and without quota, on average, non-comparable. Therefore, to estimate the causal effect of the quota, we exploit the rule-based assignment of female quotas across GPs which creates discontinuities in the probability of quota assignment as a function of demographic characteristics of GPs; we consequently employ a fuzzy regression discontinuity design (RDD) method for identification. We find that, on average, gender quotas in 2015 increase the probability of allocation of toilets to eligible households in 2016–17 by 12–28% (depending on bandwidth). The average effect, however, is imprecisely estimated. We then show that the average noisy effect masks significant heterogeneity across GPs. We find that the gender quota effect is significantly larger in GPs with a higher Muslim share. For every 10 percentage points increase in Muslim population share in a GP, the quota effect increases by 3–5 percentage points. This is large considering the average probability of toilet provision in 2016–17 was 10%.

We perform a battery of tests to show that our results are not driven by pre-existing characteristics of GPs or changes in the political outcome. We examine heterogeneous treatment effects of quotas on GP population, various demographic characteristics, pre-existing stock of publicly provided goods such as schools, roads, tap water, drainage, electricity etc.,

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and allocation of toilets in the three years preceding the local elections, and find null effects on all of them except on primary schools. The main result is also robust to including GP level controls and their interactions with the quota treatment. Moreover, gender quotas had no differential effect on electing a Muslim leader in higher Muslim share GPs.[3] Additionally, to allay any concerns that variation in Muslim shares across GPs may not be exogenous, for robustness we exploit changes in GP boundaries due to a delimitation exercise before the 2015 elections. The re-drawing of GP boundaries – a purely administrative exercise – led to an exogenous change in Muslim share across GPs. We use this to estimate the heterogeneous treatment effects induced by the *change* in Muslim share, conditional on the pre-delimitation Muslim share, and find that our results continue to hold using this alternative identification strategy.

One reason for the above findings could be that gender quotas induce women to be more vocal about their preferred policy, and this "demand effect" would arguably be stronger for women with a greater preference for the policy. There is increasing evidence that the presence of female leaders in politics positively influences women's grassroots political mobilization and communication of their preferences and beliefs to the leaders (Costa, 2021; Goyal, 2020; Liu & Banaszak, 2017). We argue that a household, conditional on its preference for toilet, may register a greater demand for it under a female leader if it is headed by a woman. We show using pan-India survey datasets, that women in female-headed households (across both religions) exercise greater decision-making power within the household, enjoy greater autonomy in public participation, and participate more in the local political activities within the village. To test the "demand mechanism", therefore, we examine heterogeneity in the gender quota effect across households headed by men and women, within the same religion.[4]

We find that, women from both Hindu and Muslim female-headed households are more likely (relative to male headed households) to approach a male Sarpanch to resolve sewage and water related issues. Importantly, in presence of female quota, women from Muslim female-headed households are differentially more likely (relative to Muslim male headed households)

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to approach the Sarpanch to resolve these issues. However, there is no such pattern among Hindu female-headed households. Therefore, gender quota leads to increased *expressed demand* from Muslim female-headed households (i.e., those having stronger preference and greater autonomy to express the preference), but not from the Hindu ones. In line with this, we find that gender quota leads to a differentially large increase in the probability of toilet allocation to *Muslim* female-headed households, but not to Hindu female-headed households. Understandably, male and female headed households are different on other dimensions such as poverty, household size etc., which could also explain the result. To address this, we carry out additional robustness exercises including utilizing the household panel structure of our data. We estimate the heterogeneous treatment effects by exploiting within household variation in toilet provision between 2015–16 (pre-election) and 2016–17 (post-election) and find the same results. However, we cannot completely rule out the possibility that some other unobservable differences between Muslim and Hindu female-headed households make them more suitable candidates to receive toilets from female Sarpanches.

Our work contributes to the literature on gender quotas in politics by reconciling and explaining the mixed evidence about its effects observed across studies. Existing literature has documented that gender quotas may not be effective for public goods with minimal gender gap in preference (Clots-Figueras, 2012a; Chattopadhyay & Duflo, 2004), while others have shown that women can have different relative preference intensities based on the nature of public goods as well as for the same good depending on their social and economic characteristics (Gottlieb *et al.*, 2018; Greenberg, 2001). Our study brings together the two strands of literature by showing that the heterogeneity in women's intensity in preference, owing to cultural factors, can result in heterogeneous treatment effects of the gender quota. The pioneering study by Chattopadhyay & Duflo (2004) that led to the emergence of this literature, shows that the gender gap in preference for public goods is important in shaping the gender quota effect. They also discuss the possibility that women leaders may induce women citizens to communicate their preferences more to them, e.g., through written complaints

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to the elected leader. However, they do not find compelling evidence to show that women citizens' preferences affect the policy decisions of the women leaders. They write: "Individual women are thus not particularly more responsive to the needs of women and men in their communities. Rather, it is because their own preferences are more aligned to the preferences of women that they end up serving them better." (p. 1434) In our context, we provide suggestive evidence that women's greater expression of demand for a good can also be one of the mechanisms explaining the gender quota effect.

The remaining paper is organized as follows. We provide evidence for the gender gap in toilet preference across Hindus and Muslim households based on the prevalent cultural norms in Section 2 and discuss the policy background and main data used for analyses in Section 3. We present the identification strategy and results for overall gender quota effects in Section 4 and the heterogeneity by Muslim share in Section 5. We then discuss evidence in favor of the demand mechanism explaining the obtained results in Section 6 and conclude in Section 7.

2 Rural Sanitation and Preference for Toilets in India

2.1 Rural Sanitation

We use toilet allocation as our primary policy outcome. Sanitation policies have been around in India since 1986. These programs, however, were largely unsuccessful in achieving the desired reduction in open defecation rates. The proportion of households having toilets in India stood at 47% in 2011 (Census 2011); in rural India it was 30.7%. The state of Uttar Pradesh, the context of our study, fared worse than the national average, with only 22% of rural households having access to toilets in 2011. We focus on rural India, since according to Census 2011, 92% of households without access to a toilet or latrine were rural. Similarly, the Indian Human Development Survey (IHDS) 2011–12 reports that 90% of households practicing open defecation live in rural areas. The toilet allocation policy received significant boost in recent periods, after a new government was formed in India in 2014, which we utilize

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for our analysis. We discuss this in Section 3.1.

2.2 Women's Relative Preference for Toilets

We first argue that women have a greater preference for household toilets than men (Coffey *et al.*, 2014; Khanna & Das, 2016; Stopnitzky, 2017). This is motivated by the observation that notions of shame and dignity are associated more strongly with women, especially in rural India, leading to their lower preference for defecating in the open. Men, on the other hand, consider defecating in the open a sign of masculinity. Moreover, women face a greater risk of being harassed or attacked while defecating in the open (Hossain *et al.*, 2022). Additionally, menstruating women may prefer to use toilets due to hygiene concerns. Coffey *et al.* (2014) show direct evidence of gender gap in toilet preference by capturing within household variation in open defecation rates in the SQUAT survey (2014). The SQUAT survey was conducted in rural areas of north Indian states, namely Bihar, Uttar Pradesh, Haryana, Madhya Pradesh, and Rajasthan, and captures information on household ownership and individual use of toilet. These states are culturally similar in their gender attitudes and include the state that we study.[5] They find that among the households owning a toilet, men are twice as likely to defecate in the open than women. We validate this using the SQUAT survey data. We find that women are 9 percentage points more likely to use a toilet, conditional on the household owning one, even within the same household (Appendix Table C1: Panel A, columns (1) and (2)). Coffey *et al.* (2014) argue that this is consistent with a lower revealed preference for open defecation among women. This finding could be consistent with other reasons as well. For example, since men are more likely to work outside home, it can lead to more open defecation by them. We also find that among the households that do not own toilets, women report having higher preference for toilets than men (Appendix Table C1: Panel A, columns (3), (4) and (5)).[6] We discuss the details on SQUAT data, variable construction, and results in Appendix Section A.1.

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2.3 Gender Gap in Preferences across Religions: Role of Cultural Norms

Having established the existence of gender gap in the preference for household toilets, we now argue that this gender gap is larger among Muslims than Hindus. This is motivated by existing evidence in the literature. Coffey *et al.* (2017) discuss why open defecation is more common in rural India relative to Sub-saharan African countries, despite higher per capita incomes, education and water availability.[7] They show that cultural norms affecting notions of 'purity and pollution' among Hindus, associated with defecating within a house, are an important factor for higher open defecation rates in India. These concerns, among both Hindu men and women, increase their alignment on the issue of toilet ownership and partially narrow the gender gap in preferences among Hindus. Such purity concerns are absent among Muslims. Additionally, Muslim women face greater restrictions on mobility than Hindus, resulting in greater demand for household toilets from them. We find evidence of this claim in the 2015–16 National Family Health Survey (NFHS) that asked women if they can visit various public spaces in their village on their own. Muslim women are 7 percentage points less likely to say yes, relative to Hindu women (Appendix Table C3: Panel B, column (2)). Consequently, they are more likely to own and use toilets than Hindus, despite being poorer than Hindus on average. Coffey *et al.* (2017) report that only 4% of rural Hindu households used inexpensive pit toilets, compared to 15% of rural Muslim households. Data from the 2015–16 National Family and Health Survey (NFHS) also show that Muslims are 21% less likely than Hindus to defecate in the open. In fact, Geruso & Spears (2018) argue that differential sanitation practices of Muslims and Hindus can explain the longstanding puzzle in India that, "[...]Muslim children are substantially more likely than Hindu children to survive to their first birthday, even though Indian Muslims have lower wealth, consumption, educational attainment, and access to state services."

We again use the SQUAT survey data to validate this. We find that among households

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owning toilets, 74% of Hindu men regularly use it, while the share is 83% for Hindu women. For Muslims, the shares are 82% and 96% among men and women, respectively. Hence, the gender gap in usage of toilets is 50% higher among Muslims compared to Hindus. Appendix Table C1: Panel B, columns (1) and (2) show that the result holds even after controlling for the household's economic status and within the same village or household. Among households not owning toilets, 44% and 46% of Hindu men and women report toilets to be their most preferred good, while the numbers are 41% and 51% for Muslim men and women, respectively. Therefore, the gender gap in toilet preference is also significantly higher among Muslims (Appendix Table C1: Panel B, columns (3)–(5)). Appendix Section A.2 discusses these results in detail.

3 Background and Data

3.1 Clean India Mission

In the most recent efforts to increase households' access to toilets, in October 2014 the newly formed central government in India launched the Swachh Bharat Mission (SBM) or the 'Clean India Mission'. The government allocated substantially higher resources under the SBM relative to a similar program under the previous government due to the high policy priority accorded to it. We specifically examine construction of household toilets in rural Uttar Pradesh (UP). The main thrust under the rural component of SBM was to provide a subsidy of ₹12,000 towards the construction of household toilets.[8] All households below the poverty line (BPL) were eligible for this subsidy. Among the above poverty line (APL) households, those belonging to Scheduled Castes, Scheduled Tribes, small and marginal farmers, landless laborers with homestead, physically handicapped, and women-headed households were also eligible.

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3.1.1 Toilet Construction Data

We compile household level toilet construction data using detailed information available on the official website of SBM maintained by the Ministry of Drinking Water and Sanitation (MDWS), Government of India.[9] We scraped the data during March to April 2018. The website provided a full list of households living in each GP and for each household, indicated whether it had a toilet at the time of the baseline survey, carried out in 2012. It subsequently tracked each household from 2013–14 onwards and indicated whether it had a toilet at the end of each financial year. This information is given along with the name of the household head, the name of the parent or spouse of the head, and certain household characteristics such as whether it was Below Poverty Line (BPL), landless, and headed by a woman etc.

For our purpose, we focus on toilet construction in the year 2016–17, the year following the Sarpanch elections in UP. This was the first financial year post the GP elections in UP. Moreover, in 2017, the ruling party at the center (the BJP) won the UP state election. The state government subsequently heavily pushed the SBM scheme in the state, especially in areas where construction was lagging, since the target set by the central government was to reach 100% toilet access by the end of 2019. Hence, toilet construction during 2016–17 is more likely to reflect political will of the Sarpanch and household demand in the GP, rather than being driven by the policy priorities of the state government. In rural Uttar Pradesh, the proportion of households having a toilet increased from 32% in 2012–2013 to 37.5% in 2015–2016 and further to over 44% in 2016–2017 (i.e., by 7 percentage points in just one year post the launch of SBM). The summary statistics for toilet provision, conditional on not having a toilet at the end of the previous financial year, are provided in Appendix Table C4: Panel B. The pace of toilet construction increased post the launch of SBM with the largest increase of 12.6% occurring in 2016-17, the year immediately after the Gram Panchayat election in UP.

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Data Quality: We provide evidence that the administrative data are of reasonable quality for the period we consider. In an independent study, Gupta *et al.* (2020) resurveyed a subset of households from the SQUAT survey of 2014 in 2018, to examine toilet construction under SBM and changes in open defecation during 2014–18.[10] They report that 73% of households in rural UP had toilets in October, 2018.[11] According to the administrative data, 64% of households in rural UP had toilets at the end of 2017–18, i.e., by March 2018. These figures are comparable. Moreover, they also observe a large drop (26 percentage points) in open defecation during 2014–18 and find that "nearly the entire change in open defecation between 2014 and 2018 comes from increases in latrine ownership, rather than from changes in behavior." This validates the toilet construction figures reported in administrative data up to 2018 and confirms that toilet construction under SBM in the initial years led to perceptible changes in the practice of open defecation in these states. Moreover, Singh *et al.* (2021) show using SBM data that toilet provision in the initial years of the program (2015-2016) led to significant reduction in under-5 stunting in rural India. This indicates that toilet allocation as reflected in the SBM data is correlated with real improvements in health outcomes.[12]

3.2 Gram Panchayat Elections

The village councils or Gram Panchayats (GPs) are the lowest tier of the local governance in rural India. The council in a GP consists of elected members and a president or head of the council, who is known as the Sarpanch of the GP. In UP the Sarpanch is elected directly following a presidential system. The council is responsible for provision of local publicly provided goods such as hand pumps, toilets, local roads etc. and employment under public works. Moreover, the Sarpanch within the council enjoys substantial executive power in deciding the priorities of the council over provision of publicly provided goods and its overall expenditure pattern (Besley *et al.*, 2004; Das *et al.*, 2017).

The GPs played a pivotal role in the implementation of the SBM program. While the program implementation was monitored at the district level by the District Magistrate,

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the primary implementing agency at the village level was the GP. The council oversaw the identification of potential beneficiaries, fund flow and maintenance of records, mobilization of demand for construction of toilets, actual construction of toilets, and social audits.

Election data: The GP Sarpanch elections in Uttar Pradesh were held during November–December 2015. These elections are subject to affirmative action policies or quotas for three caste (or ethnic) groups (OBCs, SCs and STs) as well as for women. This is known as the "reservation policy." Detailed results for the 2015 GP Sarpanch elections come from the State Election Commission of Uttar Pradesh. We focus on the 2015 election since this is the first election after the current central government came into power in 2014 and gave the sanitation campaign a major push by launching the SBM program in October 2014. There are over $59,000$ GPs in UP and about $470,000$ candidates contested the Sarpanch elections in 2015. The election data contain information on candidate characteristics, such as their gender, caste etc., as well as the votes received by each of them. The election data also provide the assignment of caste and gender quotas for the Sarpanch position in each GP.

Due to the delimitation of constituencies of GPs based on the 2011 census, the assignment of SC, ST and OBC reservation, as well as female reservation for the 2015 elections was done afresh, disregarding the allotment status in the previous elections.[13] Unlike some other states, the allocation of reservation positions across GPs in UP is not randomized but is based on a deterministic algorithm. The rules of reservation assignment are discussed below.

Caste Reservation: The reservation status of GPs in the 2015 Sarpanch elections in UP was decided separately for each block. A block is the higher administrative tier relative to a GP. The proportion of Sarpanch positions reserved for a caste group in a block equals the rural population share of that caste group in the block. The number of GPs to be reserved is then determined by rounding off the calculated share. The GPs within a block were first arranged in the descending order of the ST population share of the GPs. Then the top ranked GPs were reserved for ST, where the number of reserved GPs is given by the rounded off

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number calculated in the previous step. Then the remaining GPs within the block were arranged in descending order of their SC population shares. Again, the top ranked GPs were selected to be reserved for SC. For the remaining GPs within the block, OBC reservation status was decided using the same procedure. We refer to GPs not reserved for any of the caste groups as caste-unreserved GPs.

Female Reservation: At least a third of Sarpanch positions allocated to every caste group in each block are reserved for females of that caste. For this, the GPs reserved for a caste group are ranked in descending order of the population share of that caste. Top one-third of these GPs, rounded off to the higher integer, are reserved for women. The caste-unreserved GPs in a block are listed in descending order of the "general" category population in the GPs and again, the top one-third are reserved for women (from any caste). The "general" category refers to upper castes, i.e., those who are not STs, SCs, or OBCs. We implement the caste and gender reservation algorithms using the GP level caste wise population data provided by the UP State Election Commission (SEC), the same data that the government officials used for quota assignment purposes. We can predict the allotment of female reservation in almost 98 percent of GPs using the above algorithm.

3.3 Caste and Other Covariates

The caste group wise population (for General caste, OBCs, SCs, and STs) at the GP level, used as the basis for reservation allocation by the state government, were accessed from the UP State Election Commission. We use the Census 2011 village population and amenities data and the mapping between census villages and the GPs prepared by the Ministry of Panchayati Raj to construct GP level covariates on a host of demographic and development variables.[14]

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3.4 Religion Identification

Hindus and Muslims are the primary religious groups in UP comprising more than 99% of the rural population (Census, 2011). However, data on election results and toilet construction do not provide information on religion. Therefore, we identify religion using names of household heads and candidates in village elections. We can do this because the names of Hindus and Muslims in India are linguistically distinct. For instance, Bhalotra *et al.* (2014) infers religion of electoral candidates in India from their names. In line with this, we manually classify religion of candidates as Muslim or non-Muslim (which we refer to as Hindu) based on their names and the names of their parent or spouse.

To infer the religious affiliation of the households, we use the name of the household head and their father's or spouse's name. There are about 25 million households in rural Uttar Pradesh. Therefore, manual classification is not feasible. To overcome this, we use a new and highly accurate algorithm proposed by Chaturvedi & Chaturvedi (2023) who infer religion from names using character- sequence-based machine learning models.[15] Within a random sample of manually annotated 20,000 households in rural UP, the model correctly identifies over 97% of true Hindus as well as true Muslims. Another test of the algorithm's accuracy is the correlation between the predicted Muslim household share from our data and that from the Census 2011 at the sub-district or *tehsil* level—the lowest geographic unit for which religion-wise population figures are available. Figure C2 shows the relation between Muslim population share (Census 2011) and the Muslim household share estimated by the algorithm at the tehsil level. The correlation between the two is 0.978. This shows that the model predicts religious affiliation of households very well in our data.

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4 Overall Effect of Gender Quota

4.1 Identification

We use a fuzzy regression discontinuity design (RDD) strategy to find the causal effect of female reservation on construction of toilets (the ATE). As discussed above, no less than one-third of the Sarpanch positions reserved for each caste group in every block are reserved for women of that caste. The procedure for allotment of gender quota described in Section 3.2 creates discontinuities in the mapping between a GP's rank in the ordered list and its female reservation status. We exploit these discontinuities to estimate the overall gender quota effect since the gender reservation status of the GPs is essentially randomized around the discontinuity for each caste group in each block. We define the running variable $X^c_{g,b}$ in GP g in block b reserved for caste group c as follows:

$$X^c_{g,b} = \frac{Pop^c_{g,b} - \overline{Pop}^c_b}{\sigma^c}$$

$Pop^c_{g,b}$ is population share of caste c in GP g in block b for $c \in \{SC, ST, OBC\}$, while it is the total general category population in the GP when $c \in \{Unreserved\}$. $\overline{Pop}^c_b$ is the threshold for caste c in block b. It is the mean of the lowest $Pop^c_{g,b}$ at which the GP Sarpanch position should be reserved for women and the $Pop^c_{g,b}$ of the next GP in the ranking for caste c. The denominator (σ^c) is the standard deviation of $Pop^c_{g,b}$ across the entire state. We follow this procedure to generate the running variable for GPs reserved for each caste category, i.e., ST, SC, OBC, as well as for the unreserved GPs. Since each GP can only be reserved for one caste or remain unreserved, we get a unique running variable for each GP. We therefore refer to $X^c_{g,b}$ simply as X_g.

McCrary test: We test for manipulation at the threshold using the test proposed by McCrary (2008). This tests for the null hypothesis that the density of the running variable

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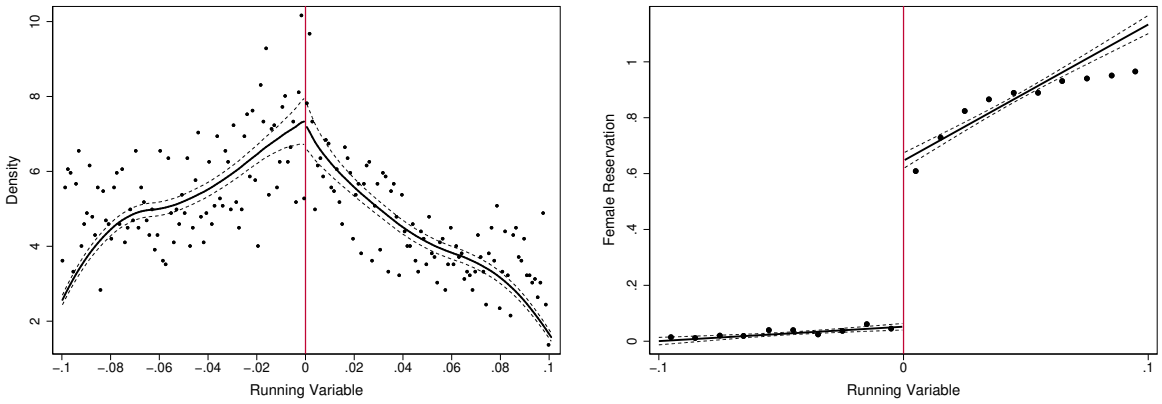
Figure 1. RD plots: McCrary Plot and First Stage



(a) Overall McCrary Test **(b)** Overall First Stage

that defines the assignment is continuous at the cutoff, against the alternative of a jump in the density at that point, which can reflect manipulation in treatment assignment. Figure 1a shows that there is indeed no jump in the density of the running variable at the threshold.

Balance test: We perform balance tests on a large number of covariates around the discontinuity to test the validity of the RDD approach. These include a range of development indicators compiled using the Census 2011 village amenities dataset as well as various demographic variables. The results for the balance tests performed for three bandwidths of 0.1, 0.075 and 0.05 are shown in columns (1)–(3) of Appendix Table C5: Panel A. We find that there is no discontinuity in these covariates at the threshold across all bandwidths.

We also test for discontinuity in toilet provision at the threshold in the three years before the 2015 Sarpanch elections. Appendix Table C5: Panel B reports the results. The proportions of uncovered households that were provided toilet in 2013–14, 2014–15 and 2015–16 do not change discontinuously at the threshold. These results demonstrate that the female reservation assignment is exogenous at the threshold of the running variable.

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4.2 Estimation Strategy

We use X_g to predict the treatment status of a GP in the first stage. The treatment variable (Q_g) equals one if the GP has a quota for women. The assignment variable (A_g) takes the value one if the female reservation algorithm predicts that the Sarpanch position should be reserved for a woman ($X_g \geq 0$) and 0 otherwise ($X_g < 0$). We restrict the sample to GPs where the running variable is near zero, and run the following regression:

$$Y_{h,g} = \alpha_0 + \tau Q_g + f_1(X_g) + f_2(X_g) * A_g + u_g \tag{1}$$

where $Y_{h,g}$ is a dummy variable that takes value one if household h in GP g received a toilet during 2016–2017 under the SBM, and zero otherwise. $f_1()$ and $f_2()$ are the control functions defined over the running variable on the two sides of threshold. In the baseline specification, we use linear control functions and show that our results are robust to quadratic functional form. Our sample is the set of eligible households for the program which did not have toilets at the end of 2015–2016. The treatment variable (Q_g) is instrumented with the assignment variable (A_g) in the following first stage equation:

$$Q_g = \beta_0 + \gamma A_g + v_1(X_g) + v_2(X_g) * A_g + \epsilon_g \tag{2}$$

where $v_1()$ and $v_2()$ are the control functions, analogous to the previous equation. We use the number of eligible households in a GP as weights in our regressions to give equal consideration to all GPs in the household level data. The standard method used to select bandwidths in RDD is the one proposed by Calonico *et al.* (2014). However, the "CCT bandwidth" is 0.475 for the second stage, which is particularly wide as it includes about 64% of GPs in the state. We, therefore, estimate the results using three narrower bandwidths—0.100, 0.075 and 0.050 and cluster the standard errors at the GP level. The bandwidths correspond to 17.3%, 13.6% and 9.9% of GPs, respectively. Also, since we test for heterogeneity in the

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gender quota effects across different sub-samples, fixed bandwidths maintain consistency and facilitate comparison of estimates across specifications.

4.3 Results

We graphically show the first stage results in Figure 1b. It shows that the likelihood of female reservation is zero on the left of the threshold and jumps to about 0.6 at the threshold. Appendix Table C6 reports the regression result for specification (2). It shows that we have a strong first stage across all three bandwidths and the estimated discontinuity in the probability of female reservation is in the range of 0.53–0.59.

Table 1—Gender Quota Effect

	HH Received Toilet in 2016–17		
	(1)	(2)	(3)
Female reservation	0.0133	0.0200	0.0285
	(0.0156)	(0.0186)	(0.0235)
Mean dep. var. in control GPs	0.107	0.107	0.100
Observations	2,470,191	1,962,725	1,457,226
Number of GPs	9,179	7,234	5,277
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if the household received a toilet in 2016-17, and zero otherwise. The sample includes eligible households, i.e., those that did not have toilet at the end of 2015-16 and were eligible to receive toilet under the SBM scheme. The polynomial order is 1. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

We now discuss the second stage results using the household level data. Table 1 reports the average effect of female reservation on the probability that any household without a toilet received one in 2016–2017. The coefficients at the three bandwidths—0.1 (column (1)), 0.075 (column (2)) and 0.05 (column (3))—are 0.0133 (95% CI: $[-0.0174, 0.0439]$), 0.0200 (CI: $[-0.0165, 0.0564]$) and 0.0285 (CI: $[-0.0175, 0.0745]$) respectively, i.e., they are all positive. The effect sizes vary from 13% (in column (1)) to 28% (in column (3)) of the mean allocation probability in the control GPs (i.e., GPs with $X_{g,b} < 0$), i.e., they are moderate

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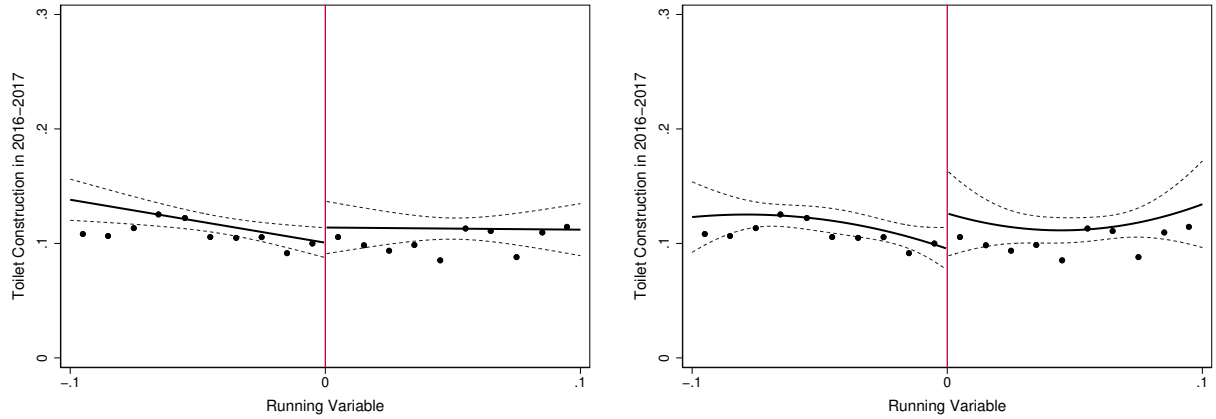
Figure 2. RD plots: Second Stage



(a) Linear Polynomial **(b)** Quadratic Polynomial

Notes: The Figure reports RD plots based on the specification in Equation (1) using linear (Panel a) and quadratic (Panel b) polynomials fitted on either side of the threshold within a bandwidth of 0.100.

in size. But all the coefficients are statistically insignificant due to large standard errors. The average treatment effect is, therefore, positive but noisy. Figure 2 shows the second stage results graphically with linear and quadratic control functions in Panels (a) and (b), respectively. They also demonstrate that there is no statistically significant average effect of the gender quota. In the following section we show that the average result subsumes significant heterogeneity across GPs.

5 Heterogeneity in Gender Quota Effect

We now test for differential effects of gender quota across GPs with different Muslim household shares. Since the gender gap in preference for toilets is larger among Muslims due to cultural norms, a higher share of Muslims in a GP implies that the average gender gap in preference in the GP would be larger. We establish this using the SQUAT survey data. As before, we examine toilet usage among households having a toilet and reported preference ranking of toilets among households without toilets.

Table 2 reports the results. We find that women in villages with a higher share of Muslims

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Table 2—SQUAT Survey: Muslim Share and Gender Gap in Toilet Preference

	Toilet Usage		Toilet Preference		
			Top	Top 2	Top 3
	(1)	(2)	(3)	(4)	(5)
Female	0.083***	0.083***	0.033	-0.012	0.00107
	(0.00882)	(0.00687)	(0.0327)	(0.0305)	(0.0273)
Female * Muslim share	0.125***	0.133***	0.336	0.434**	0.378**
	(0.0434)	(0.0367)	(0.220)	(0.200)	(0.184)
Mean Dep. Var.	0.80	0.80	0.46	0.65	0.78
Observations	7,752	7,738	1,477	1,477	1,477
Fixed Effect	Village	HH	Village	Village	Village

Notes: The dependent variable for columns (1) and (2) is a dummy that takes value one if the individual uses the latrines for defecation. The samples in the first two columns only include households which had latrines. The dependent variable in each of the columns (3), (4), and (5) is a dummy that takes value one if the respondent reports a toilet as being the topmost, top two or top three priorities, respectively, if money was not a constraint, from a list of assets that the household does not have. The samples in the last three columns only include households without latrines. 'Female' is an indicator variable that takes a value of one for women and zero otherwise. 'Muslim share' is the proportion of Muslims in a village estimated from the SQUAT survey. Columns (1) and (3)-(5) have village fixed effects and control for household level assets and household's main source of income. Column (2) has household level fixed effect. The number of observations in columns (1) and (2) differ with village and household fixed effects as single member households are dropped from the analysis. Robust standard errors clustered at village level are in parentheses. The SQUAT states are Haryana, Uttar Pradesh, Madhya Pradesh, Bihar, and Rajasthan. The survey was conducted in December 2013-April 2014. *** p<0.01, ** p<0.05, * p<0.1.

use toilets significantly more relative to men (columns (1) and (2)), and are also more likely to report a higher preference for toilets relative to men in households that do not own one (columns (3)-(5)). Since variation in the Muslim share across villages could be correlated with observable and unobservable characteristics of households, we include household fixed effects (FE) in column (2) and find consistent results. The larger gender gap in toilet preference in villages with a higher Muslim share can both be due to Muslims having higher gender gap in preference (as described in Section 2.3) and Hindu households living in high Muslim share villages exhibiting greater gender gap in preferences than those living in lower Muslim share villages. We test the latter point in Appendix Table C7 by running the same specification used

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in Table 2 with an added triple interaction term Muslim * Female * Muslim share and other two-way interactions. We find suggestive evidence for it. The coefficient for the interaction term Female * Muslim share, which reflects the gender gap among Hindus in villages with varying Muslim shares, is always positive and is statistically significant in column (2).

This can happen because of multiple reasons. Hindu women are more likely to learn the beneficial effects of in-house toilets in villages with higher Muslim shares. The possibility of greater behavioral spillovers is also higher in such cases. A larger gender gap in preference for toilets in a GP implies that the female leader in the GP may be induced to prioritize toilet provision. We therefore expect that the gender quota effect would be larger in GPs with a higher Muslim household share.

There is significant variation in Muslim shares across GPs in Uttar Pradesh (Appendix Figure C3). This allows us to estimate the heterogeneous effects of gender quota using the *difference-in-discontinuities* approach proposed by Grembi *et al.* (2016). We specifically estimate the following equation:

$$\begin{aligned} Y_{h,g} &= \alpha_0 + \tau Q_g + \alpha_1 X_g + \alpha_2 X_g * A_g \\ &+ S_g * [\theta_0 + \rho Q_g + \theta_1 X_g + \theta_2 X_g * A_g] + u_g \end{aligned} \tag{3}$$

where S_g is the Muslim share of GP g. We instrument the treatment variable Q_g and its interaction with S_g using the assignment variable A_g and its interaction with S_g. Specifically, we have two first stage equations given by:

$$Q_g = \beta_0 + \gamma A_g + \beta_1 X_g + \beta_2 X_g * A_g + S_g * [\delta_0 + \lambda A_g + \delta_1 X_g + \delta_2 X_g * A_g] + \epsilon_g \tag{4}$$

$$Q_g * S_g = \beta_0^{'} + \gamma^{'} A_g + \beta_1^{'} X_g + \beta_2^{'} X_g * A_g + S_g * [\delta_0^{'} + \lambda^{'} A_g + \delta_1^{'} X_g + \delta_2^{'} X_g * A_g] + \epsilon_g^{'} \tag{5}$$

For the instrumental variable method to work in this case we need the monotonicity assumption to hold for both Q_g and $Q_g * S_g$. We instrument Q_g and $Q_g * S_g$ with A_g and

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$A_g * S_g$. Since A_g positively predicts Q_g, as Appendix Table C6 shows, $A_g * S_g$ would also positively predict $Q_g * S_g$. Hence, we expect the monotonicity condition to hold.

Table 3—Heterogeneous Impact of Female Reservation on Toilet Allocation

	HH Received Toilet 2016–17		
	(1)	(2)	(3)
Female reservation	-0.0204	-0.0272	-0.0357
	(0.0205)	(0.0243)	(0.0305)
Female reservation*Muslim share	0.292**	0.403**	0.539***
	(0.139)	(0.164)	(0.200)
Muslim share	-0.0277	-0.0745	-0.110*
	(0.0460)	(0.0520)	(0.0618)
Mean dep. var. in control GPs	0.107	0.107	0.100
Observations	2,470,191	1,962,725	1,457,226
Number of GPs	9,179	7,234	5,277
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if the household received a toilet in 2016–17, and zero otherwise. The sample includes eligible households, i.e., those that did not have toilet at the end of 2015–16 and were eligible to receive toilet under the SBM scheme. The polynomial order is 1. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Results: Appendix Table C8 reports the first stage results. Panel A shows the results for Q_g and Panel B for $Q_g * S_g$. The results show that the first stage holds for A_g and $Q_g * S_g$. Table 3 reports the second stage results. We find that the estimate of ρ is positive, large and statistically significant for all the three bandwidths. Moreover, the magnitude of the estimate increases from 0.29 (95% CI: $[0.020, 0.564]$) to 0.54 (CI: $[0.147, 0.931]$) as the bandwidth narrows. The estimate of τ is small, negative, and statistically insignificant. This implies that there is no effect of female reservation on toilet allocation in GPs with zero Muslim share. However, for every 10% increase in Muslim share, the gender quota effect on probability of toilet allocation increases by $0.03 - 0.05$. Given that the average probability of allocation is around 0.1 in GPs without female reservation, the result shows that there is a large heterogeneity in the gender quota effect. Appendix Table C9 shows that the result

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is robust to using quadratic control function. The coefficient estimates are similar across the three bandwidths, and similar in magnitude to column (3) of Table 3.[16] This suggests potential underfitting using linear control function at bandwidths of 0.1 and 0.075. The result is also robust across a range of bandwidths (Appendix Figure C6). Consistent with the mixed evidence found in the literature, we find that even within a state, the gender quota can produce highly heterogeneous effects.

Pre-existing Characteristics of GPs: GPs with varying Muslim shares could also differ along other pre-existing characteristics. Appendix Table C10 shows, for example, that GPs with higher Muslim shares have higher population, more primary schools, roads and irrigation and lower electricity. If these correlations are different across GPs with and without female reservation, then it could be a potential confounder. To check this possibility, we estimate equation (3) using baseline characteristics of GPs and allocation of toilets in years preceding the election as dependent variables. Appendix Table C11 reports the results. We find that barring the presence of primary schools within 5 km, in none of the other 16 outcomes we find a significant result across the three bandwidths. Additionally, we also interact Female Reservation with GP level controls in specification (3) and report the results in Appendix Table C12. We find that our result remains robust. It is therefore unlikely that pre-existing observable characteristics of the GPs drive the heterogeneous treatment effects by Muslim household share.

Supply Side Explanation: It is also possible that female reservation causes a differentially higher probability of a Muslim getting elected as Sarpanch in GPs with a higher Muslim share. This can also drive the heterogeneous treatment effects by directly affecting leader's preference. Appendix Table C13 reports the results of estimating equation (3) using a dummy for Muslim Sarpanch as the dependent variable. We find that the estimate of ρ is positive, but small and statistically insignificant for all the bandwidths. Appendix Figure C7 shows the local polynomial relationship between Muslim share and probability of electing Muslim Sarpanch

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for GPs with and without female reservation. We do not find any statistical difference between the two relationships. We, therefore, do not find evidence of a significant difference in this supply side characteristic of GPs to be driving the observed heterogeneity. Additionally, we empirically examine in Appendix Section B.4 whether Muslim women Sarpanches are more likely to allocate toilets by looking at close elections between Hindu and Muslim candidates in GPs with and without female reservation (in a difference-in-discontinuity framework). We find null effects. Though Muslim women leaders winning close elections (against Hindu women) in a reserved seat may not be fully comparable to those who do not face close elections, the result suggests that the difference in preferences between Muslim and Hindu female leaders is unlikely to be the only factor driving our main results.

Robustness Exercise: Muslim share of a GP could be correlated with some local unobservable characteristics of the GPs (such as level of trust, or collective action etc.) which could potentially confound the heterogeneous treatment effects. To address this concern, we carry out a robustness exercise by exploiting changes in Muslim shares induced by redrawing of the GP boundaries before the 2015 elections. The State Election Commission redrew the GP boundaries – known as the delimitation exercise – to balance populations across GPs. This would induce changes in the Muslim share of GP. We map census villages to pre- and post-delimitation GPs to construct the change in Muslim household share for each GP. We argue that this *change*, conditional on the pre-delimitation Muslim share, is potentially exogenous. We find that our results are robust to this estimation strategy as well (Appendix Table C14). The details are presented in Appendix Section B.1.

Placebo Exercise: We consider the possibility that the heterogeneous treatment effects could result from differentially higher ability women getting elected due to the quota in GPs with a higher Muslim share. Hence, it is ability of the leader rather than preference intensity of women that drives the result. We use implementation of National Rural Employment Guarantee Scheme (NREGS) – a large public workfare program – as our outcome variable

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and find no statistically significant heterogeneous treatment effects (Appendix Table C15). This indicates that ability is unlikely to be the primary explanation of our results. Appendix Section B.3 discusses the result in detail.

However, increase in Muslim population share, even if exogenous, may cause unobservable changes in the GP's political dynamics which could potentially make gender quotas more effective. We are unable to rule out this possibility completely.

6 The Demand Channel

Thus far, we have shown that differential preferences across GPs with varying Muslim shares (directly through presence of more Muslims and indirectly through similar preferences of Hindu households) can shape the heterogeneous effects of gender quota. This can happen either due to greater expression of toilet preference by GP residents or differential preferences of GP leaders by gender. In this section, we focus on the first mechanism – the "demand channel". Appendix Section B.4 discusses the second "supply mechanism" and as mentioned in the previous section, does not find supportive evidence.

We posit that women voters' demands may be heard more in the presence of female leaders (Iyer & Mani, 2019). Hence, for policies that they have a stronger preference for, they may express demand for it more vigorously in the presence of female leaders (Chattopadhyay & Duflo, 2004; Beaman *et al.*, 2007), resulting in higher treatment effect. We examine if this "demand effect" is important in our context. Toilet is a household level good, even though preference for it differs starkly across men and women within the household. Therefore, conditional on the preferences of household members, we argue, the expression of demand for toilets to the local leader may depend on who is at the helm of the household. For instance, Coffey *et al.* (2014) discuss that young women are likely to have the largest demand for toilet use but have the least decision making power in a household. Therefore, the capacity or willingness of a household to register its demand may depend on the relative bargaining

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power, autonomy, and political participation of women in the household, which may be higher in female-headed households. We verify this claim using two nationally representative survey datasets, namely the 2015–16 National Family Health Survey (NFHS-4) and Rural Economic and Demographic Survey 2006 (REDS) data.

We show that women from female-headed households are significantly more likely to make household decisions on their own, venture out alone in public spaces, attend village meetings, and participate in activities of the political parties. We report the results in Appendix Table C3: Panel A and discuss them in Appendix Section A.3. Moreover, this difference between female and male headed households exists across both Hindus and Muslims (Appendix Table C3: Panel B). Overall, female-headed households comprise 4% of our estimation sample. Appendix Table C10 shows that proportion of female headed households in GPs is not significantly correlated with Muslim share. Additionally, Appendix Figure C8 plots the share of female headed households across Muslim shares. We again see no correlation between the two. Therefore, examination of toilet allocation to them may shed light on the proposed demand mechanism. We test (i) if female headed households with stronger preference for toilets among women express greater demand for sanitation to female leaders, and (ii) whether provision of household toilets is consistent with the demand.

Direct Evidence of Demand: We use the REDS 2006 survey data, a nationally representative survey of rural households in India, to examine demand. It asks respondents whether they faced any issue regarding various public services, such as water supply, sewage, roads etc., and if yes, whether they approached any government representative for grievance redressal. We do not have a direct measure of complaints regarding sanitation in the REDS data. As a proxy, we focus on women respondents who faced water supply and sewage related issues. We use water related complaints since the availability of water is complementary to the usage of sanitation facilities. Similarly, sewage related complaints might be related to both raw sewage and refer to toilets. We define a dummy variable that takes value one if the

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respondent, conditional on facing issues on those two items, approached the Sarpanch, and zero otherwise. While the variable we construct is imperfect, it can potentially capture the expression of preference for sanitation to the Sarpanch.

This is similar to the written complaints that Chattopadhyay & Duflo (2004) examined to test if female reservation led to more women expressing their preferences. Additionally, REDS is a pan-India survey and predates the SBM policy, allowing us to show evidence on expression of demand for all of rural India. We regress the dummy variable defined above on whether the respondent belongs to a female-headed household, the female reservation status of the village as of 2006, and their interaction, along with individual and village level controls. We estimate the results separately for Hindu and Muslim households since the gender gap in preference is higher among the latter.

Table 4—Water and Sewage Related Complaints by Female Headed Households

	Complaint was expressed to Sarpanch			
	All India		North India	
	Hindu HH (1)	Muslim HH (2)	Hindu HH (3)	Muslim HH (4)
Female reservation (2006)	0.0317	-0.0837***	-0.0163	-0.0848**
	(0.0282)	(0.0286)	(0.0386)	(0.0356)
Female reservation (2006) * Female Headed Household	-0.0473	0.242**	-0.100	0.266*
	(0.0554)	(0.116)	(0.0680)	(0.143)
Female Headed Household	0.114***	0.0822	0.197***	0.154**
	(0.0324)	(0.0576)	(0.0473)	(0.0613)
Mean dep. var.	0.48	0.29	0.39	0.24
Fixed Effect	District	District	District	District
Observations	7,957	440	3,492	271
R-squared	0.164	0.276	0.121	0.158

Notes: The data source is the Rural Economic and Demographic Survey (REDS) 2006. Only women respondents older than 18 years are considered. Columns (1) and (3) has the Hindu sample, while columns (2) and (4) the Muslim sample. First two columns consider the sample from all of rural India, while the last two columns restrict the sample to north Indian states of Bihar, Uttar Pradesh, Haryana, Rajasthan and Madhya Pradesh. The dependent variable is a dummy that takes value one if the respondent had approached the Sarpanch for grievance redressal with regard to water and sewage related issues, and zero otherwise, conditional on facing issues on any of the two items. Female reservation (2006) is a dummy indicating female reservation status in the surveyed GPs as of 2006. Controls include age, age squared and landholding of respondents, and total number of households and SC population share of the village. Standard errors clustered at village level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 4: columns (1) and (2) show the results separately for Hindu and Muslim women,

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respectively. Our coefficient of interest is the interaction between female reservation and female-headed Household (the second row). We find that women from Muslim female-headed households are significantly more likely (by 24 percentage points (95% CI: $[0.015, 0.470]$) to complain about water and sewage under a female Sarpanch (column (2), row (2)). There is, however, no such pattern for Hindu female-headed households (column (1), row (2)). The coefficient for the interaction term is small and statistically insignificant. Additionally, the results remain the same when we restrict the sample to the same north Indian states as the SQUAT survey (columns (3) and (4)). Both Hindu and Muslim female-headed households are not differentially likely to complain under a male Sarpanch, even though Muslim women have higher intensity of preference. This suggests that Muslim female-headed households have more unexpressed complaints under a male leader, possibly due to their belief that they might not be heard by a male Sarpanch. Presence of a female Sarpanch, therefore, induces them to complain more.

Estimation of Demand Effect: We now test whether the allocation of toilets to female-headed households follows patterns consistent with the results above. We estimate the following difference-in-discontinuity specification:

$$\begin{aligned} Y_{h,g} &= \alpha_0 + \tau Q_g + \alpha_1 X_g + \alpha_2 X_g * A_g \\ &+ F_{h,g} * [\theta_0 + \rho Q_g + \theta_1 X_g + \theta_2 X_g * A_g] + u_g \end{aligned} \tag{6}$$

where $F_{h,g}$ is a dummy variable that takes a value of one if a household is headed by a woman and zero otherwise. Our main coefficient of interest is ρ. As before, Q_g is instrumented with A_g in the following first stage equations:

$$Q_g = \beta_0 + \gamma A_g + \beta_1 X_g + \beta_2 X_g * A_g + F_{h,g} * [\delta_0 + \lambda A_g + \delta_1 X_g + \delta_2 X_g * A_g] + \epsilon_g \tag{7}$$

$$Q_g * F_{h,g} = \beta_0^{'} + \gamma^{'} A_g + \beta_1^{'} X_g + \beta_2^{'} X_g * A_g + F_{h,g} * [\delta_0^{'} + \lambda^{'} A_g + \delta_1^{'} X_g + \delta_2^{'} X_g * A_g] + \epsilon_g^{'} \tag{8}$$

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Importantly, we estimate equation (6) for Hindu and Muslim households separately. Let ρ^H and ρ^I be the estimates of ρ for the Hindu (H) and Muslim (I) household samples, respectively. We hypothesize that if expression of demand is important in shaping the gender quota effect, then ρ^H would be zero and ρ^I would be positive.

Results: We estimate the above regression on the subset of Above Poverty Line (APL) households for whom we know the gender of the household head.[17] Table 5 columns (1)–(3) report the results for Hindu households for the three bandwidths, while columns (4)–(6) report the same for Muslim households. Consistent with the evidence on expressed demand, we find that the estimate of ρ^H is small and statistically insignificant (columns (1)-(3), row (2)), while ρ^I is positive, large, and statistically significant (columns (4)-(6), row (2)) across all bandwidths for Muslim households. The estimate of ρ^I ranges from 0.31 (95% CI: $[0.095, 0.515]$) – 0.35 (CI: $[0.108, 0.582]$), which is a large increase in probability of allocation, given that the mean allocation probability for Muslim households is about 0.15. It is also robust across a range of bandwidths (Appendix Figure C9). Table 5 also reports the number of female headed households in each specification. It ranges from about 4,000 to over 57,000. Hence, the results in the table are less likely to be sensitive to small sample size related issues.

The coefficient for female-headed household in row (3) is positive in Table 4, but negative in Table 5. This suggests that male Sarpanches discount demand expressed by FHHs, which rationalizes their change in behavior under female Sarpanches.[18]

Alternative Interpretations: Alternatively, it is possible that FHHs are economically weaker. Appendix Table C16: Panel A, using the NFHS data, shows that on average female headed households are poorer, have less education, and are older with fewer family members. This is consistent with the literature that finds that in female headed households either the male spouse of the household head is no longer alive or has separated/divorced (Lingam, 1994). We also find that across the country 66% and 4% of female and male heads, respectively, are widowed, separated or divorced while 32% of the female heads are currently married. Though

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Table 5—Impact of Gender Quota on Female Headed Households

	Dep. Var.: HH Received Toilet					
	Hindu Households			**Muslim Households**		
	(1)	(2)	(3)	(4)	(5)	(6)
Female reservation	0.0195	0.0290	0.0436	0.0324	0.0591	0.116*
	(0.0196)	(0.0233)	(0.0291)	(0.0432)	(0.0514)	(0.0636)
Female reservation * Female Headed Household	0.0591	0.0619	0.0368	0.305***	0.345***	0.326**
	(0.0482)	(0.0575)	(0.0700)	(0.107)	(0.121)	(0.141)
Female Headed Household	-0.0332*	-0.0326	-0.0183	-0.0821***	-0.0767***	-0.0492
	(0.0193)	(0.0218)	(0.0241)	(0.0265)	(0.0273)	(0.0300)
Mean dep. var. in control GPs	0.12	0.12	0.11	0.15	0.15	0.14
Observations	1,414,073	1,116,043	832,032	188,489	148,547	107,177
Number of Female Headed Households	57,647	45,548	32,449	7,686	6,146	4,099
Number of GPs	8,858	6,981	5,096	7,878	6,189	4,521
Polynomial order	1	1	1	1	1	1
Bandwidth	0.100	.075	0.050	0.100	0.075	0.050

Notes: The dependent variable is a dummy that takes value one if a household received toilet in 2016–17, and zero otherwise. The sample includes above poverty line (APL) eligible households, i.e., those that did not have toilet at the end of 2015-16 and were eligible to receive toilet under the SBM scheme. The first and last three columns are Hindu and Muslim households, respectively. Female Headed Household is a dummy that takes value one if the household head is female and is zero otherwise. The polynomial order is 1. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

the NFHS does not capture migration, existing evidence from the Indian Human Development Survey (2004-05) shows that 97% of currently married female heads have migrant spouses (Mahajan, 2019). It is then possible that these households may need the toilet subsidy more, resulting in greater allocation by the Sarpanch. Notably, we restrict the analyses of female headed households to only those who are above poverty line since the gender of the household head is only available for them. To that extent, some of these concerns are reduced, though we cannot fully rule them out. Also, we find that the coefficients for FHHs (θ_0 in equation 6) across all the columns in Table 5 are negative and large in magnitude, implying that those households are *less likely* to receive toilets relative to male headed households under Sarpanches without female quota. Moreover, Muslim FHHs are on average less poor and more educated than Hindu ones (Appendix Table C16: Panel B), further indicating that economic condition of FHHs cannot justify the observed allocation patterns. Additionally, we do not find any evidence of differential effect of female reservation on Scheduled Caste and

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Scheduled Tribe households (poorer castes in India), among the APL households (Appendix Table C17). Hence, female Sarpanches do not seem to be differentially focused on poorer households.

Second, it could also be that female Sarpanches are more responsive to demands made by FHHs, or they are more concerned about FHHs than male Sarpanches. However, in such a case, we should expect the estimates of ρ to be similar across Hindu and Muslim female-headed households, which we don't find. A third possibility is that Muslim FHHs are different in some unobservable characteristics that make them more suitable beneficiaries of SBM, which could explain our findings. To address this concern, we carry out a robustness exercise. In this alternative specification, we take advantage of the panel nature of data and compare changes in allocation *within* a household during 2015–16 (pre-election period) and 2016–2017 (post period) using household fixed effects. This allows us to control for all the time-invariant observable and unobservable characteristics of a household. Appendix Table C18 shows that our result remains robust as the estimated coefficient on $Post *$ Female reservation $*$ Female Headed Household is positive and significant for Muslim households (columns (4)-(6)) while it is insignificant for Hindu households (columns (1)-(3)). We discuss the estimation strategy details in Appendix Section B.2. However, given the differences between male and female headed households along various household characteristics, it is difficult to completely rule out all the possible explanations of the result. It is therefore appropriate to treat our results as suggestive in nature. Finally, FHHs are a small fraction of the GP population, which ensures that differential allocation to them is unlikely to be motivated by electoral concerns of female Sarpanches. We independently test whether female leaders' preferences on their own explain our results. As discussed in Appendix Section B.4, we find suggestive evidence against this.

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7 Conclusion

Quotas for women in politics, despite their worldwide popularity and adoption, have been found to have a highly heterogeneous impact on women's substantive representation across countries and policies. We examine provision of household toilets in a large North Indian state and find that the electoral gender quota exhibits substantial heterogeneity in its effect even within a given political context and for the same publicly provided (private) good for which women have stronger preference than men. We argue that an important reason for this is that women's relative intensity of preference for a policy may itself vary. We show that cultural and social factors shape the gender gap in preference for toilets across Hindus and Muslims in India. This results in heterogeneity in women's relative intensity of preference for toilets across villages with different religious composition. We show that this preference heterogeneity in fact drives the heterogeneous treatment effects on toilet provision. We additionally find that women from female headed households, who complain more about sewage and water related issues to female leaders, are also more likely to receive toilets from them. This suggests that the demand effect can be important in shaping the effect of gender quotas.

Our result, therefore, highlights the important role that female voters can play in shaping the gender quota effect. Our findings suggest that policies that empower women voters and encourage them to participate in political processes and express their preferences and beliefs more freely can make gender quotas more effective. There is, for instance, some evidence that enhancing women's social capital through self-help groups facilitates their collective action and political participation (Desai & Joshi, 2014). Such policies, therefore, can potentially aid existing gender quota policies in improving women's substantive representation.

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Notes

[1]The information about gender quotas in politics across countries is available from the IDEA Gender Quotas Database: https://www.idea.int/data-tools/data/gender-quotas/country-overview.

[2]The gender gap in stated preference for toilets is even higher among Muslims.

[3]GPs with a higher Muslim shares are more likely to elect Muslim leaders, but not differentially so in presence of female quotas.

[4]We do not examine heterogeneous treatment effects across Hindu and Muslim households, as Sarpanches could have electoral motives to allocate more toilets to the majority group—Hindus. Alternatively, Muslim Sarpanches may allocate more toilets to Muslim households due to own-group favoritism.

[5]The dataset is publicly available at https://riceinstitute.org/data/2014-and-2018-rural-sanitation-surveys/.

[6]The survey asks about preference for having toilets for the household. The exact question asked is: "Now I will read out a list of items that you don't have. I want to ask you, which are the three things that you would want to have for your household first. Imagine that there is no lack of money to buy them." Therefore, the estimated gender gap in preference is inclusive of any considerations that men may have about the women members' preferences in the household. Additionally, it could also be that male preferences vary by presence of young daughters in the households. We check for this possibility in Appendix Table C2 and find that men's preference is insensitive to the presence of daughters.

[7]In 2015, the proportion of rural Indian population defecating in the open was 43% while this figure was 32% for rural Africa and the global average in rural areas was 20%. The figures in rural parts of comparable economies like Bangladesh stood at 1.7% and China at less than 1% (World Development Indicators).

[8]The SBM promotes construction of twin-pit structures for toilets so that the feces decompose by themselves, and no frequent manual cleaning of fecal sludge is required. This subsidy, on an average, is sufficient to cover all costs related to a twin-pit toilets in India. However, if a household wanted it could build a better-quality toilet. The subsidy amount would remain unchanged.

[9]The official website of SBM is `https://swachhbharatmission.gov.in/`.

[10]The 2018 resurvey was done in Rajasthan, Madhya Pradesh, Uttar Pradesh, and Bihar.

[11]This is an estimate for the entire rural UP arrived at using census population weights. The average estimate for the four states in the survey is 71%.

[12]Hossain *et al.* (2022) show that the correlation between district level toilet coverage in the administrative data for 2015–16 with toilet ownership reported by households in NFHS-4, also conducted in 2015–16, is fairly high (0.70). We plot the proportion of households in a district with toilet access from the two data

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sources for the state of Uttar Pradesh in Appendix Figure C1 and demonstrate a similarly high association between them. There is a small difference in coverage periods for the two data sources. The administrative data gives coverage at the end of fiscal year 2015–16 while NFHS reports an average over 2015–16. Thus, the official data are likely to report slightly higher coverage, on average.

[13]This was in accordance with the 10th amendment of Uttar Pradesh Panchayat Raj (Reservation and Allotment of Seats and Offices) Rules 1994.

[14]We access the mapping from the website www.lgdirectory.gov.in.

[15]Character sequence based models have the advantage that they can exploit differences in linguistic origins of the two religions, and hence can also classify unseen names with high accuracy. For our purpose, we use a deep learning architecture which combines convolutional neural network with long short-term memory recurrent neural network.

[16]Appendix Figure C4 shows the second stage estimates graphically, using both linear and quadratic control function. In both the specifications, we find that there is no effect of female reservation on toilet allocation in GPs with zero Muslim share but a large positive impact when the share is equal to one. In addition, Appendix Figure C5 shows the corresponding results by restricting the sample of GPs to those having Muslim share above and below the threshold t, where $t \in \{20\%, 30\%, 40\%, 50\%\}$ using both linear and quadratic polynomial specifications. We again find broadly consistent results.

[17]All Below Poverty Line (BPL) households are eligible for SBM, and hence, the data do not record the gender of the household head for them.

[18]The positive coefficients of ρ_I in Table 5 columns (4)–(6) can be interpreted as resulting from changes in Muslim women's inherent preference for toilets in response to female reservation, rather than a change in their expression of demand. This could be an alternative mechanism of the demand effect. However, the positive coefficient of Muslim FHH in Table 4 (columns (2) and (4), row 3) indicates higher priority of toilets among them even under a male leadership. Therefore, change in inherent preference of toilets is unlikely to be the primary driver of our results.

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Appendix

For Online Publication

A Preferences for Toilets: Evidence from Survey Data

A.1 Gender Gap in Preference for Toilets

We use the SQUAT survey to analyze preference for toilets. First, we examine how likely a person is to use a toilet conditional on the household owning one. In Appendix Table C1, columns (1) and (2), we look at the probability of using a toilet for defecation, by gender, conditional on household owning a toilet using village and household fixed effects respectively. Results in Panel A, columns (1) and (2), show that women are more likely to use in-home toilets. For the set of households that do not own a toilet, the survey asks a randomly chosen member of the household about his or her top three priorities for the household from a list of assets that the household does not possess, if money is not a constraint. We create three indicator variables that capture whether the respondent states toilet ownership to be among the top-most, one of top two or top three priorities. We test whether respondent gender matters for the responses. The results are reported in columns (3), (4), (5) of the same table. We find that women are more 5% points more likely to state that their top-most priority is a toilet for the house (column (3)). The estimates for the other two specifications are also positive but are statistically insignificant. Overall, our analysis demonstrates that, consistent with existing evidence, women have a stronger preference for a household toilet than men.[19]

A.2 Gender Gap in Preference between Hindus and Muslims

In panel B of Appendix Table C1, we examine the heterogeneity in the gender gap in preference for toilets across Hindus and Muslims, using the SQUAT survey. Columns (1) and (2) show that conditional on having a toilet, Muslim households are more likely to use a toilet

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than Hindu households. Moreover, the usage gap between Muslim women and men is almost double of the gap between Hindu women and men. Focusing on the households that do not yet own a toilet, columns (3), (4) and (5) show that relative to Muslim men, Muslim women are 22–24% more likely to report that it is among the top two or three priorities for them. The gender gap in reported preference among Hindus is positive but statistically insignificant. These results show that overall, the gender gap in preference for toilets is much higher for Muslims relative to Hindus.

A.3 Demand across Male and Female Headed Households

To examine female autonomy across male and female-headed households, we create two indicators using the NFHS-4 survey. The first indicator uses the set of questions on whether the woman can make any of the following decisions on her own –accessing health care, major household purchases and visits to family. The second indicator captures whether the woman is allowed alone to any of these places –market, health facility and outside village. We regress these indicators on whether a woman belongs to a household headed by a woman and report the results in Appendix Table C3 Panel A, columns (1) and (2) respectively. We control for various individual and household characteristics and village fixed effects. The results show that women residing in households headed by women are more likely to take decision and go alone to places, indicating greater decision-making power and autonomy for women in these households. We also find that female members who are not the head of the family themselves also enjoy this greater agency in female-headed households (table not reported).

Additionally, we also test if women are more politically active if they live in a woman headed household by using the REDS data. Appendix Table C3 Panel A, columns (3) and (4) show the results for whether a woman has attended any of the last four village meetings and whether a woman is actively involved with any political party, respectively. We find a positive association between female headship of the household and women's active role in community and political engagement, controlling for religion and other household and

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individual level characteristics and village fixed effects. These results show that, conditional on women having greater demand for toilets, households headed by women are more likely to express the demand for it.

In Panel B of the same table, we test if the positive relationship we found is driven by Hindus. We interact the indicator of female-headed households with the indicator of the household being Muslim. We find that the relationship (both in the NHFS-4 and REDS) is similar across Hindus and Muslims. If anything, women in female-headed Muslim households tend to enjoy greater bargaining power within the household (column (1)) and attend village meetings more (column (3)) compared to their counterparts in Hindu female-headed households.

B Additional Empirical Analysis

B.1 Causal Identification of Heterogeneous Treatment Effects

We exploit the change in GP boundaries before the 2015 village elections to get at causal identification. The GPs are typically composed of multiple census villages.[20] Since new census data became available in 2011, between two local elections in 2010 and 2015, the State Election Commission in UP rearranged the mapping between villages and GPs to bring about greater parity in population across GPs. This is known as the delimitation exercise. For each household in our data, we identify the mapping of census village (where the household lives) to the GP in 2015 (post-delimitation) and the GP in 2010 (pre-delimitation).[21] This allows us to compute the Muslim share in the pre-delimitation GPs. We explain the exercise using the diagram below.

The right-hand panel of Appendix Figure B1 shows two 2015 GPs - namely, GP 1 and GP 2 - composed of villages 1-3 and 4-6, respectively. However, as the left panel of Figure B1 shows, in 2010, villages 1, 2, and 5 were part of GP 1, and villages 2, 4 and 6 constituted GP 2. If

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the Muslim shares of villages 2 and 6 are different, then the delimitation exercise would result in changes in Muslim shares of GPs in 2015. Since we have the mapping of households to villages, as well as the mapping of villages to 2010 GPs, we can compute the Muslim shares of 2010 GPs as well. For each household h in 2015 GP g we then define:

$$\Delta S_g \equiv S_g - S_{g'}^{2010}$$

$$\Rightarrow \quad S_g \equiv S_{g'}^{2010} + \Delta S_g$$

where S_g is the Muslim share in GP g and $S_{g'}^{2010}$ is the Muslim share in the 2010 GP g' of household h. While variation in $S_{g'}^{2010}$ suffer from the same endogeneity issues as S_g, variation in ΔS_g, conditional of $S_{g'}^{2010}$, is likely to be exogenous. The officials in the State Election Commission carried out the GP delimitation exercise for the sole purpose of balancing out the population sizes across GPs. Moreover, gerrymandering is highly unlikely in this context as village elections, though important, are politically much lower stakes compared to the state and national elections. Existing evidence shows that delimitation exercises even in the state and national level elections do not suffer from any political interference (Kjelsrud *et al.*, 2020; Iyer & Reddy, 2013). We then estimate the following specification:

$$\begin{aligned} Y_{h,g} &= \alpha_0 + \tau Q_g + \alpha_1 X_g + \alpha_2 Q_g * X_g \\ &+ S_{g'}^{2010} * [\theta_0 + \rho_1 Q_g + \theta_1 X_g + \theta_2 Q_g * X_g] \\ &+ \Delta S_g * [\theta_3 + \rho_2 Q_g + \theta_4 X_g + \theta_5 Q_g * X_g] + u_g \end{aligned}$$

Our coefficient of interest is ρ_2 that estimates the differential effect of female reservation in GPs with a higher (delimitation induced) change in Muslim share. ρ_1 estimates the heterogeneous treatment effects w.r.t. the 2010 Muslim share. If the true treatment effect increases linearly with Muslim share then we should expect $\rho_2 = \rho_1$. However, if the treatment effect is convex (concave) w.r.t. Muslim share then $\rho_2 > \rho_1$ ($\rho_2 < \rho_1$).

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Appendix Table C14 reports that the estimate of ρ_2 is positive and statistically significant at all bandwidths. Therefore, exogenous increase in Muslim share in a GP increases the gender quota effect on toilet allocation. This is consistent with demand effect driving the heterogeneity. Moreover, the magnitude of ρ_2 ranges from $0.73 - 0.91$, which is larger than the estimated ρ_1, which ranges from $0.34 - 0.54$. However, we can not reject the null hypothesis that $\rho_1 = \rho_2$.

B.2 Estimation with Household Fixed Effects

Since female-headed households might differ from male-headed households in several ways, we also use a separate identification strategy in which we exploit the panel nature of the data. For this, we restrict our sample to eligible APL households that didn't own toilets at the end of 2014–15 and compare the allocation of toilets to households in reserved and unreserved GPs before and after the 2015 Sarpanch elections i.e., in the years t $\in$ {2015–16, 2016–17}. We use household fixed effects to control for all the observable and unobservable characteristics at the household level. We use a difference-in-discontinuities strategy and estimate the following specification separately for Hindu and Muslim households:

$$\begin{aligned} Y_{h,g,t} &= \alpha_{h,g} + Post_{h,g,t}[\tau Q_g + \alpha_1 X_g + \alpha_2 X_g * A_g] \\ &+ Post_{h,g,t} * F_{h,g} * [\theta_0 + \rho Q_g + \theta_1 X_g + \theta_2 X_g * A_g] + u_g \end{aligned} \tag{9}$$

where $Y_{h,g,t}$ indicates whether household h in GP g received a toilet in period t. $\alpha_{h,g}$ denotes household fixed effects. The variable $Post_{h,g,t}$ takes value 1 if t = 2016–17, and 0 otherwise. $F_{h,g}$ is a dummy variable indicating whether a household is headed by a woman. As before, our main coefficient of interest is ρ which now captures the *change in differential allocation* of toilets to female-headed households due to discontinuities in gender reservation generated by the reservation rule.[22] We instrument Q_g by A_g in the following first stage

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equations:

$$\begin{aligned} Post_{h,g,t} * Q_g &= \beta_{h,g} + Post_{h,g,t} * [\gamma A_g + \beta_1 X_g + \beta_2 X_g * A_g] \\ &+ Post_{h,g,t} * F_{h,g} * [\delta_0 + \lambda D_g + \delta_1 X_g + \delta_2 X_g * A_g] + \epsilon_g \end{aligned} \tag{10}$$

$$\begin{aligned} Post_{h,g,t} * Q_g * F_{h,g} &= \beta'_{h,g} + Post_{h,g,t} * [\gamma' A_g + \beta'_1 X_g + \beta'_2 X_g * A_g] \\ &+ Post_{h,g,t} * F_{h,g} * [\delta'_0 + \lambda' A_g + \delta'_1 X_g + \delta'_2 X_g * A_g] + \epsilon'_g \end{aligned} \tag{11}$$

Appendix Table C18, Columns (1)–(3) report the results for Hindu households while Columns (4)–(6) report the results for the sample of Muslim households at three different bandwidths. This strategy reproduces our results from Specification 6 which are reported in Table 5. We find that gender reservation does not have a differential effect on allocation to female-headed households among Hindus but differentially increases allocation to female-headed households among Muslims. The point estimates imply that the probability of allocation to female-headed households increases by 0.29–0.33 which denote a large increase in probability over the average toilet ownership of 12% in control GPs.

B.3 Placebo Test: Using an Alternative Outcome

Here we consider an alternative hypothesis, that the differential female reservation effect across GPs with different Muslim shares could be driven by differential selection of women leaders along the ability dimension. It is possible that Muslims enjoy greater collective action among themselves than Hindus. Hence, a higher share of Muslims could facilitate the electorate to select higher ability women in response to a gender quota in the GP.

To examine this, we look at implementation of the National Rural Employment Guarantee Scheme (NREGS), which is the largest expenditure head in a GP budget and constitutes an overwhelming majority of a GP's annual expenses.[23] NREGS, like the SBM, is also

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supposed to be demand driven, i.e., the extent of NREGS implementation is at least partly a function of the demand expressed for public works by households. It is also targeted at the household level. Greater provision under this scheme could potentially signal higher ability of Sarpanches to implement public projects (Bell & Mukhopadhyay, 2020). Therefore, NREGS is a good comparison policy to test whether our main results are due to differential selection of female leaders.

We estimate in Table C15 the effect of female reservation as well as its heterogeneity on per capita expenditure under NREGS. Columns (1)-(3) report the treatment effect, while columns (4)-(6) report the heterogeneous treatment effects with respect to Muslim share. We find that female reservation does not lead to greater provision under the scheme. Additionally, we do not find any significant heterogeneity in treatment effect in implementation of NREGS across GPs with different Muslim shares. The coefficients in columns (4)-(6), row (2) are positive, but statistically insignificant. Moreover, the magnitudes are also small. For every 0.1 increase in Muslim share the treatment effect increases by about 5%. The corresponding figure for toilet allocation is about 30%. Moreover, as before, we use the exogenous change in Muslim shares induced by delimitation to test its effect on NREGS expenditure. Appendix Table C19 reports the results. We again do not find any significant impact. Therefore, it is unlikely that our results are driven by differential selection of women leaders.

B.4 Supply Channel

To identify whether female leader's preference on its own explains our results, we rely on the fact that the gender gap in preference is higher among Muslims (I) than Hindus (H). Therefore, the supply effect of a Muslim female Sarpanch (relative to a Muslim male Sarpanch) would also be higher than that of a Hindu female Sarpanch (relative to a Hindu male Sarpanch). Now, generally the ATE of a Muslim and a Hindu female Sarpanch would be different, as the GPs where Muslims become Sarpanches would be very different from those with Hindu Sarpanches. However, if the religion of Sarpanches is randomly assigned (say, in

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a sample of GPs with close elections between a Hindu and a Muslim), then the GPs with Hindu and Muslim Sarpanches would have the same characteristics on average, implying that they would have the same demand effect due to gender quota. However, the overall effect of female reservation could still be different just because leaders' preferences are different.

Estimation Strategy: To test this, we select the sample of GPs where the top two candidates were a Hindu and a Muslim, and the election was close. Within this sample, the religion of the Sarpanch is effectively randomly assigned (see, for example, Meyersson, 2014). The close election regression discontinuity design is a popular method to generate random variation in the identity of elected leader (Eggers *et al.*, 2015). We use the differences-in-discontinuities strategy where the (sharp) discontinuity around close elections generates random variation in the religion of Sarpanch and the quota for women is the difference variable. We prefer this strategy over an alternative one, where we use regression discontinuity in the running variable associated with female reservation (i.e., X_g in equation (6)) to generate exogenous variation in female reservation status. This method would require us to use RDD across both religion and female reservation in the same specification. While it would generate exogenous variation in both variables, it would force us to focus on a sample of GPs that are just around the threshold value of X_g as well as experienced a close election between a Hindu and a Muslim candidate. The sample of such GPs is small and highly selective. Moreover, the practice of using RDD strategies on both differencing variables is not common. Consequently, we believe our strategy is easier to interpret and gives a more reliable estimate due to the larger sample size.

We run the following specification on the sample of close election GPs to estimate the differential effect of having a Muslim Sarpanch (relative to a Hindu Sarpanch) in female quota GPs vis-a-vis non-female quota GPs:

$$Y_{h,g} \;=\; \alpha_0 + \beta I_g + \alpha_1 V_g + \alpha_2 I_g * V_g + Q_g * [\theta_0 + \phi I_g + \theta_1 V_g + \theta_2 I_g * V_g] + u_g \qquad (12)$$

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where I_g is a dummy indicating whether the Sarpanch in GP g is Muslim. V_g is the margin of victory for a Muslim candidate, i.e., it is defined as (vote share of Muslim - vote share of Hindu). Our coefficient of interest is ϕ, which estimates the supply effect. Of course, the GPs with and without female reservation are not the same. Female reservation status is, however, almost completely determined by X_g. Hence, we show robustness of our result by controlling for X_g and allowing its slope to differ across female reserved and unreserved GPs.

Results: Appendix Table C20 reports the results for specification (12) in GPs with a close election between a Hindu and a Muslim candidate.[24] Column (1) reports the result when the absolute value of margin of victory is at most 0.1, while columns (2) and (3) report it for bandwidths 0.075 and 0.05 respectively.[25] We find that the female reservation dummy (row (1)) has a small and statistically insignificant coefficient across all the columns, implying null effect of the female quota among Hindu Sarpanches. This is consistent with our result that the gender quota effect is small in GPs with low Muslim shares, where the Sarpanches are more likely to be Hindus. For all the bandwidth specifications, we find that the difference-in-discontinuity estimate (row (3)) is also very small and statistically insignificant. Therefore, election of a female Muslim Sarpanch vis-a-vis a female Hindu Sarpanch does not change the probability of toilet construction. This implies that supply mechanism cannot explain the heterogeneous treatment effects of gender quota. The mean Muslim share in the sample of GPs used in Table C20 is 30%. Hence, the null effect is not due to lack of Muslim households in the sample of GPs with close elections. For robustness, we control for X_g and $Q_g * X_g$. The results, reported in Appendix Table C22, remain the same.

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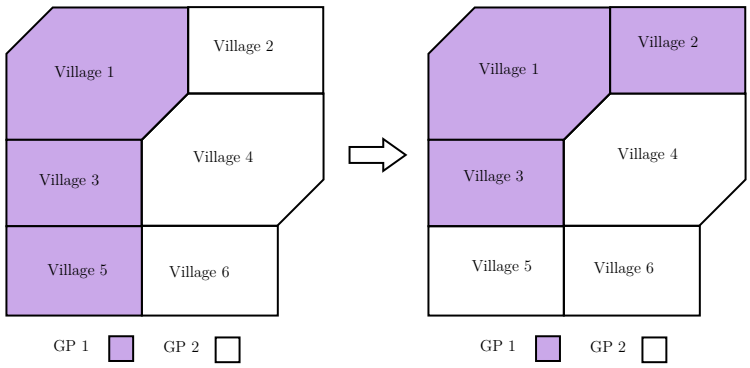
Figure B1. Delimitation Exercise of GPs



2010 GP boundaries 2015 GP boundaries

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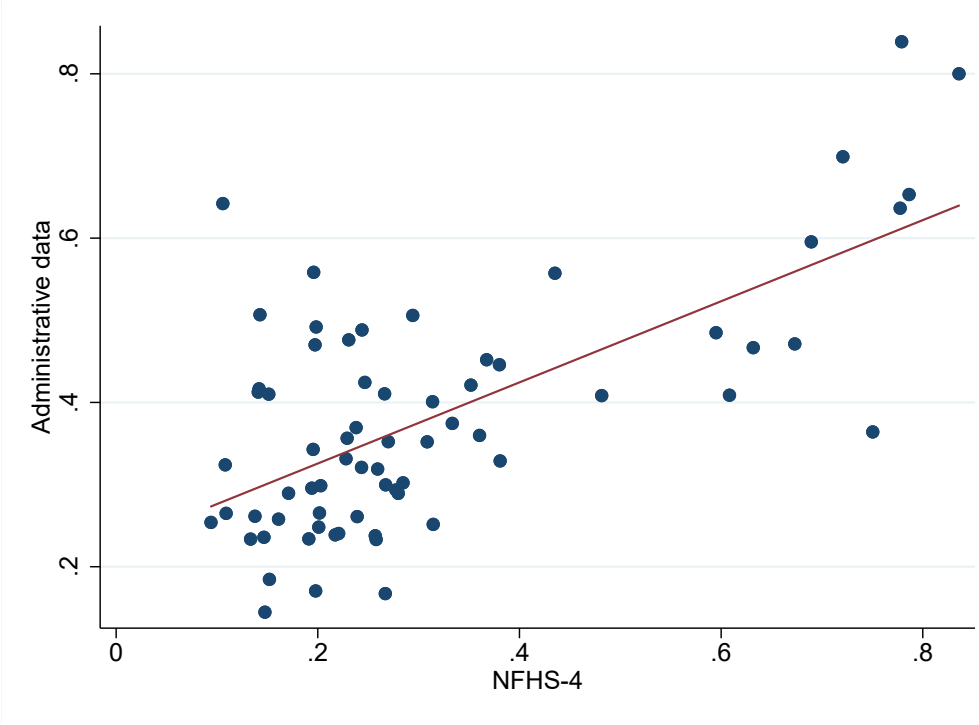
C Additional Figures and Tables

Figure C1. Toilet Coverage: Administrative Data versus Survey (2015–16)



Source: Ministry of Drinking Water and Sanitation (MDWS), India for administrative toilet data and National Family Health Survey (NFHS)-4 (2015–16) for sanitation survey data.

Notes: The figure plots the proportion of households having a toilet in the administrative data and the proportion of households having a toilet in the NFHS data for the same district. A linear relationship between the two is fitted. Correlation = 0.7.

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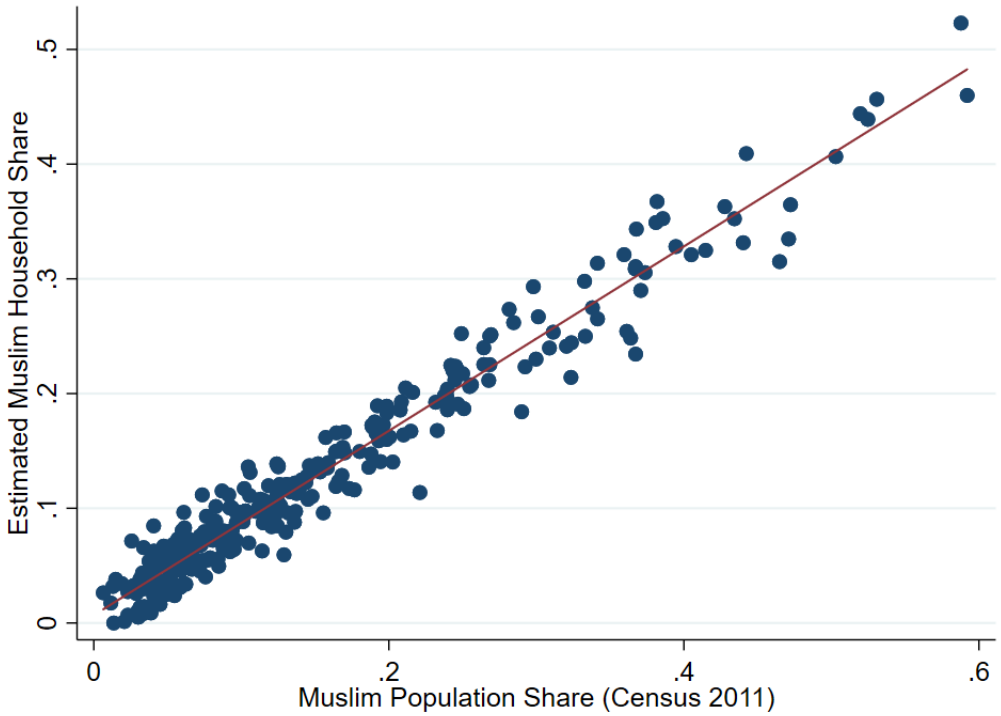
Figure C2. Comparing true and predicted Muslim population share



Notes: The true Muslim population share based on 2011 census is on the x-axis. Population share estimated by the algorithm for 312 tehsils in U.P. is on y-axis. Correlation = 0.9776.

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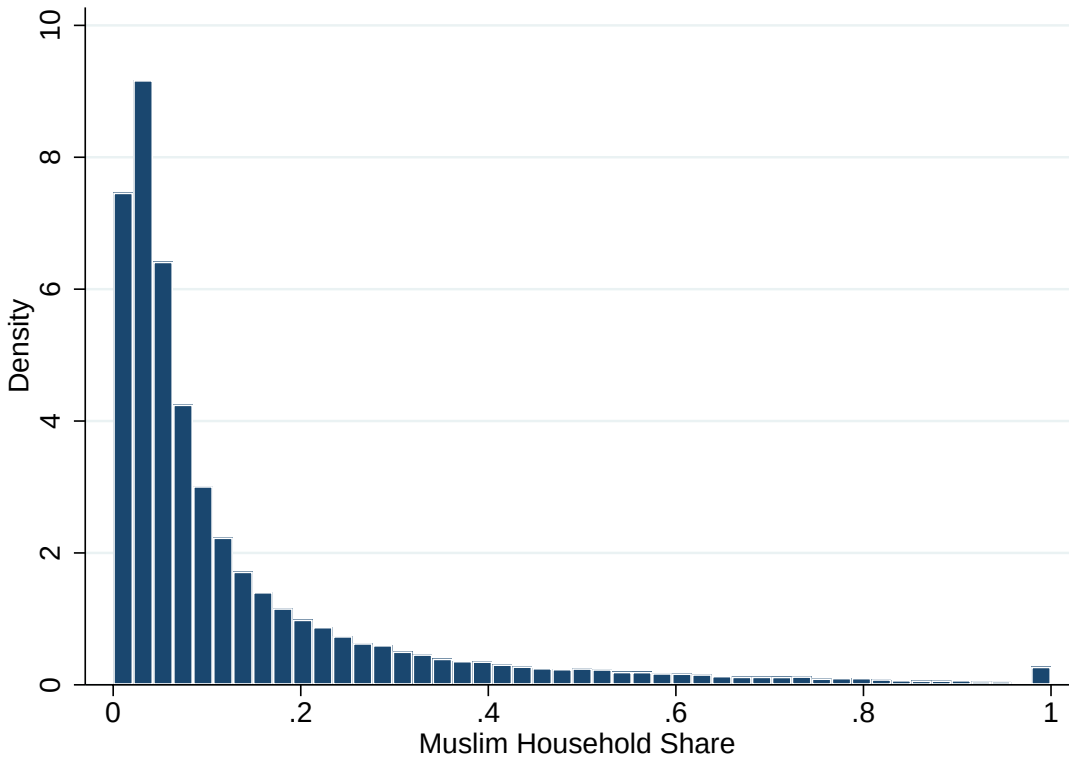
Figure C3. Density of Muslim Share



Notes: The Figure reports the distribution of Muslim household share across all the GPs in our data.

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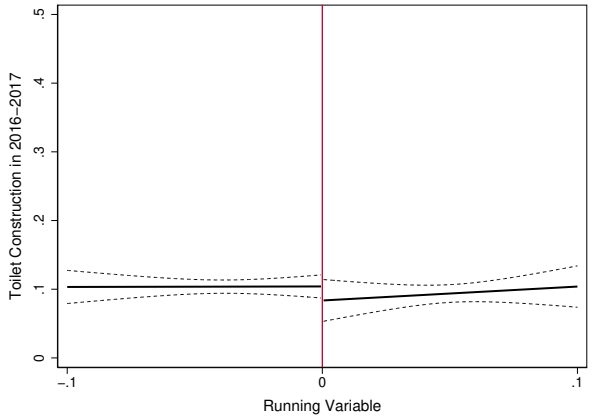
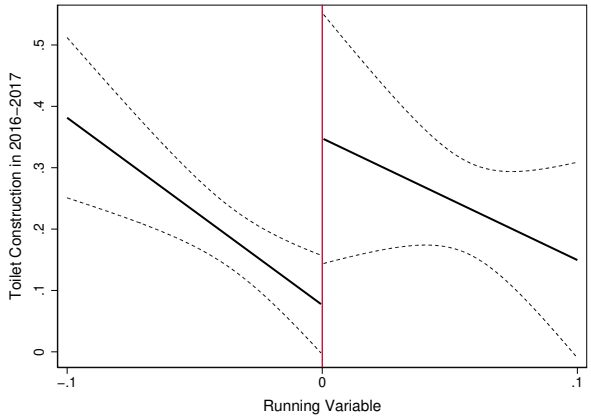
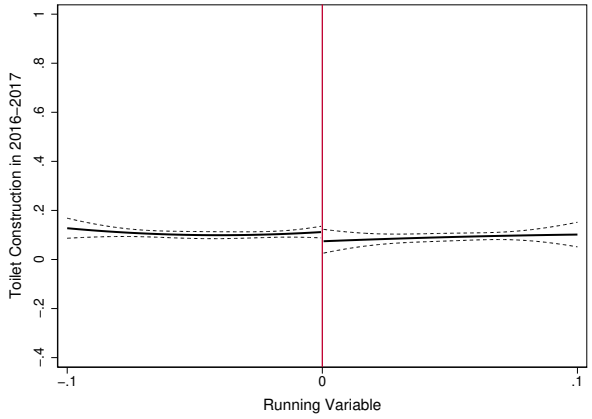
Figure C4. RD plots: Second Stage Heterogeneity



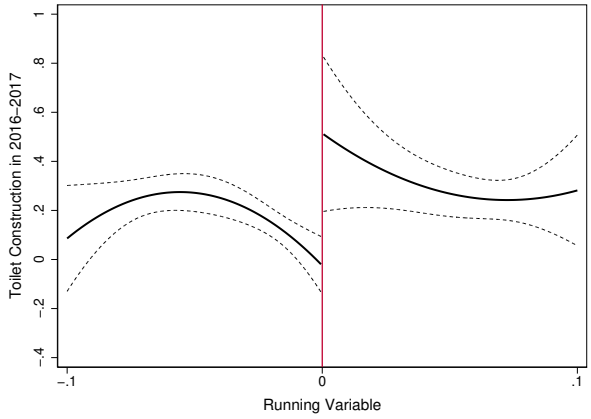
(a) Low Muslim Share (linear)



(b) High Muslim Share (linear)



(c) Low Muslim Share (quadratic)



(d) High Muslim Share (quadratic)

Notes: In Panel (a), we plot linear relationships between the running variable and toilet allocation fitted separately on the two sides of the threshold for GPs with Muslim share = 0, as estimated using Equation (3). Panel (b) plots the same relationships using the same equation for Muslim share = 1. Panels (c) and (d) show the analogous graphs using a quadratic specification. Since the plotted lines come from regression estimates, we are not able to show the binned scatter plots in this case.

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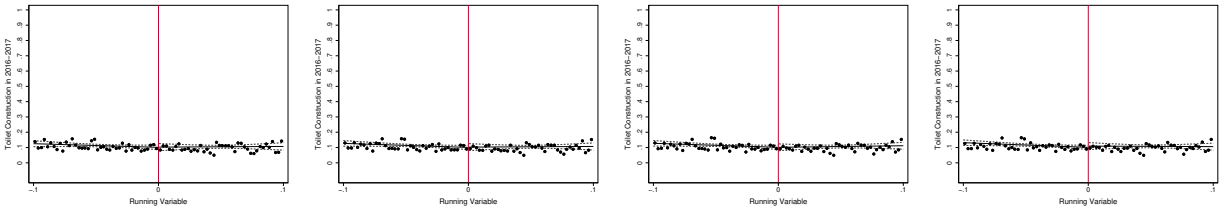
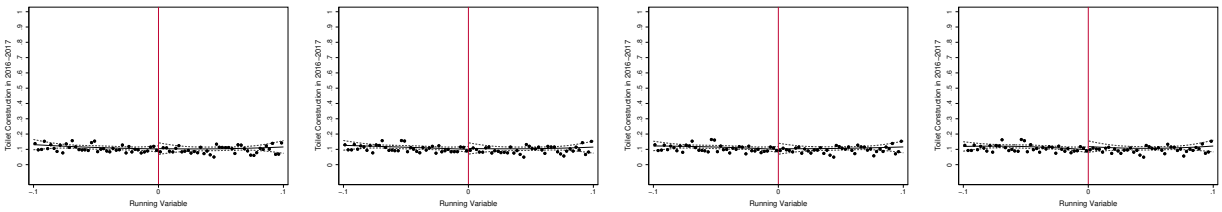
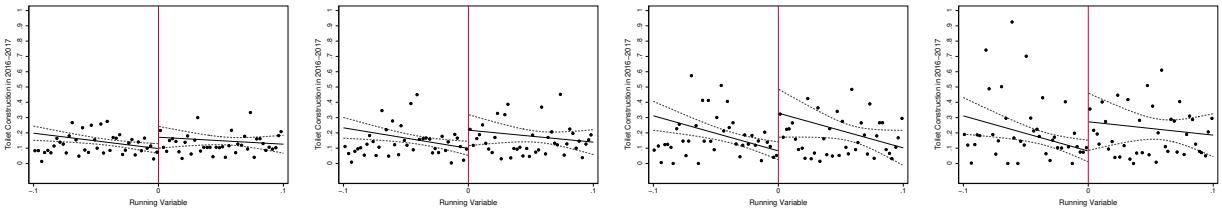
Figure C5. RD plots: Second Stage Heterogeneity



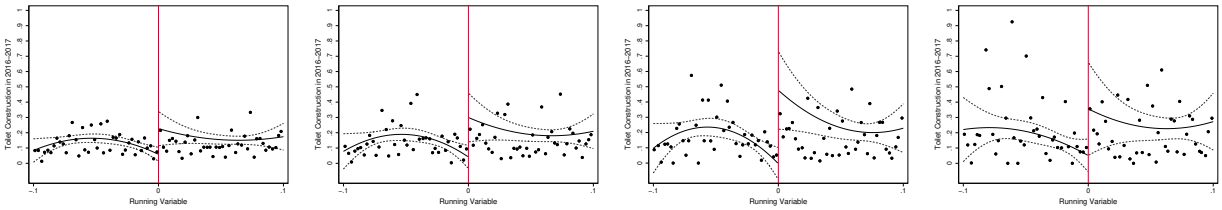
(a) Muslim Share < 20% (linear) **(b)** Muslim Share < 30% (linear) **(c)** Muslim Share < 40% (quadratic) **(d)** Muslim Share < 50% (quadratic)



(e) Muslim Share < 20% (quadratic) **(f)** Muslim Share < 30% (quadratic) **(g)** Muslim Share < 40% (quadratic) **(h)** Muslim Share < 50% (quadratic)



(i) Muslim Share ≥ 20% (linear) **(j)** Muslim Share ≥ 30% (linear) **(k)** Muslim Share ≥ 40% (linear) **(l)** Muslim Share ≥ 50% (linear)



(m) Muslim Share ≥ 20% (quadratic) **(n)** Muslim Share ≥ 30% (quadratic) **(o)** Muslim Share ≥ 40% (quadratic) **(p)** Muslim Share ≥ 50% (quadratic)

Notes: In Panels (a)–(h), we plot the relationships between the running variable and toilet allocation fitted separately on the two sides of the threshold, as estimated using Equation (1), for subsamples of GPs where the Muslim share is low. Panels (i)–(p) plot the same relationships for subsamples of GPs where the Muslim share is above a specified threshold. Panels (a)–(d) and (i)–(l) use a linear polynomial while Panels (e)–(h) and (m)–(p) show the analogous graphs using a quadratic specification.

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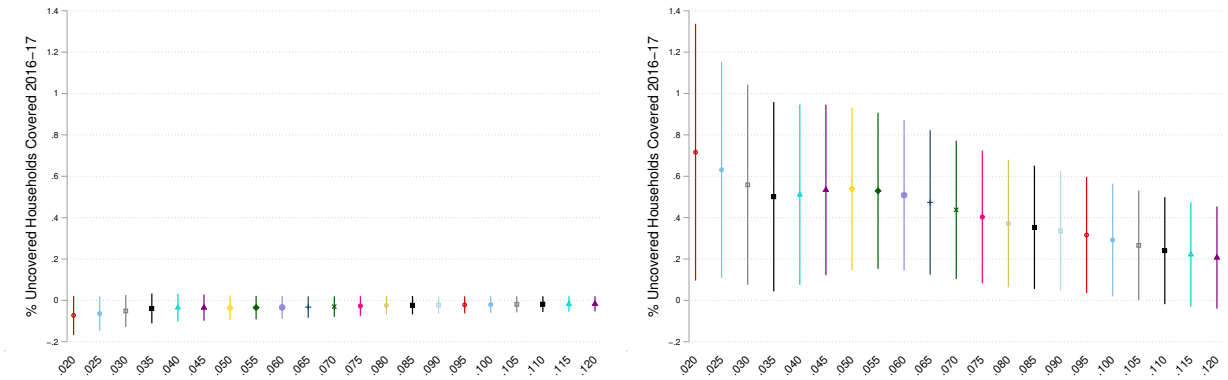
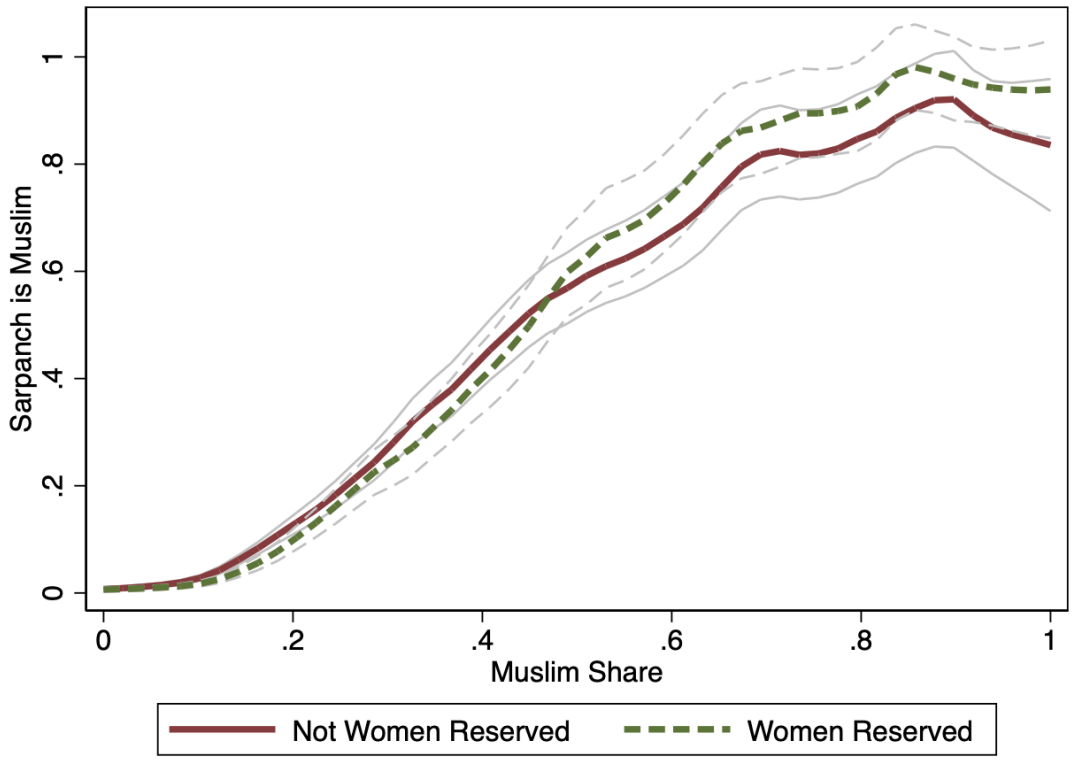
Figure C6. Coefficient Plot across Bandwidths: Second Stage Heterogeneity



(a) Female reservation **(b)** Female reservation*Muslim Share

Notes: We plot the coefficient τ (Female reservation) in Equation (3) for various bandwidths in Panel (a) and ρ (Female reservation*Muslim share) in Panel (b).

Figure C7. Muslim Sarpanches in Women Reserved and Unreserved GPs



Notes: The Figure depicts the probability of the Sarpanch being Muslim both in GPs with and without gender reservation for Muslim household share in the GPs ranging from 0 to 1 using a local polynomial estimation.

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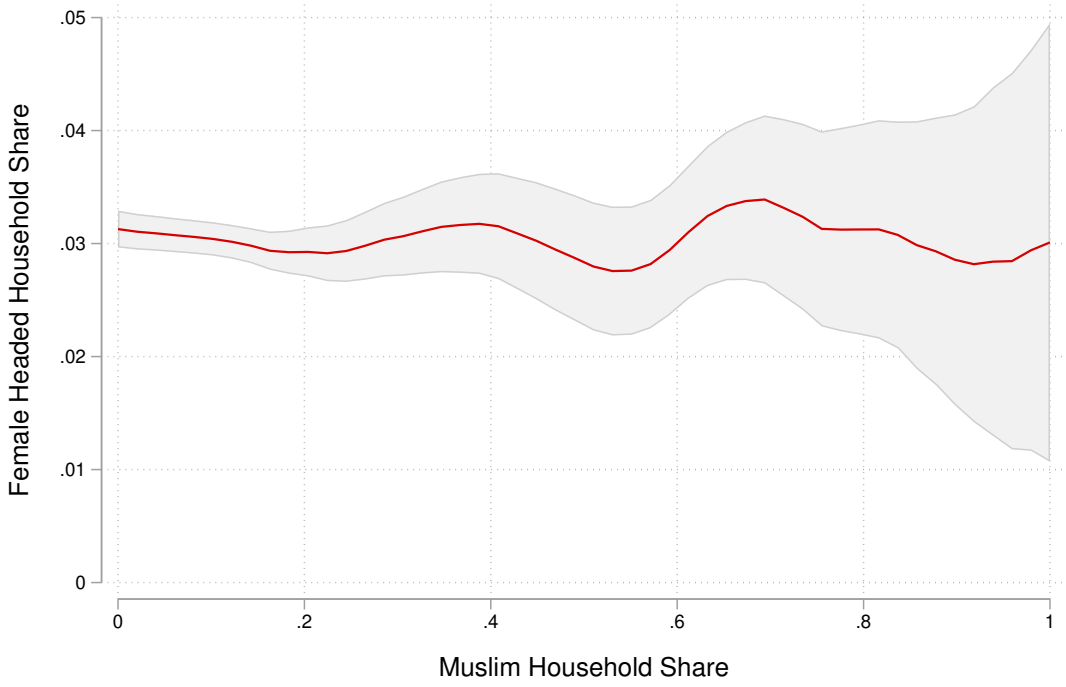
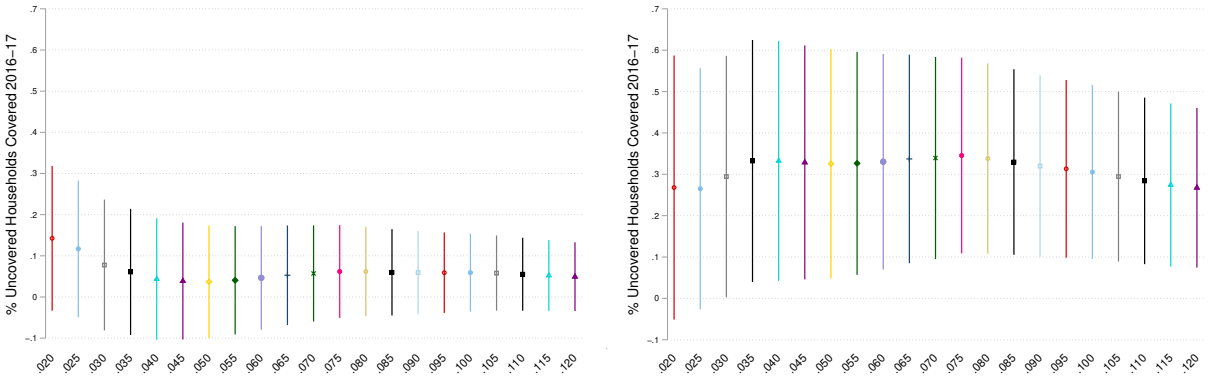
Figure C8. Female Headed Household Share and Muslim Share



Notes: The Figure plots female headed household share against Muslim household share at the GP level.

Figure C9. Coefficient Plot across Bandwidths: FHH Heterogeneity



(a) Hindu Households **(b)** Muslim Households

Notes: We plot the coefficient ρ in Equation (6) for various bandwidths in Panel (a) for Hindu households and in Panel (b) for Muslim households.

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Figure C10. Close Election: McCrary Plot

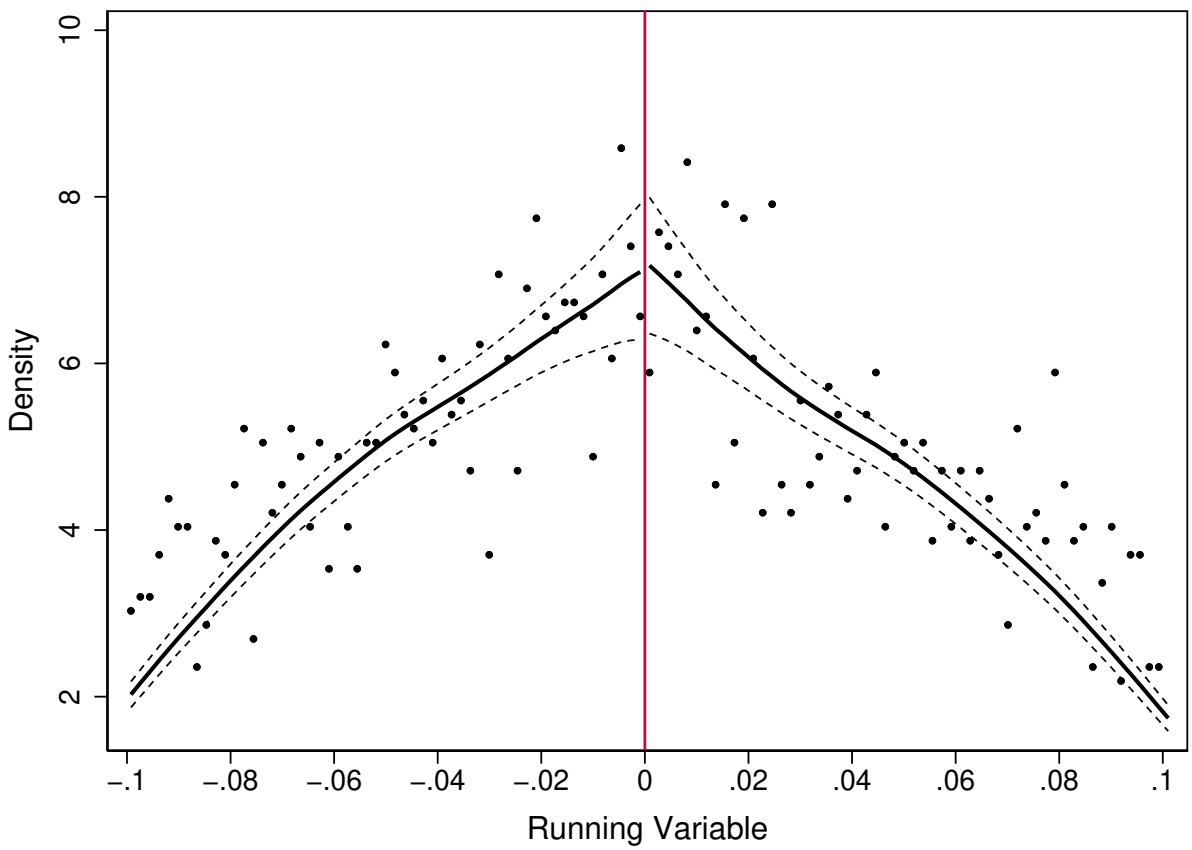

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Table C1—SQUAT Survey: Gender Gap in Toilet Preference

	Toilet Usage		Toilet Preference		
			Top	Top 2	Top 3
	(1)	(2)	(3)	(4)	(5)
	Panel A: Overall Gender Gap				
Female	0.0933***	0.0940***	0.0507*	0.0114	0.0215
	(0.00792)	(0.00620)	(0.0300)	(0.0282)	(0.0252)
	Panel B: Heterogeneity in Gender Gap				
Female	0.0889***	0.0903***	0.0456	0.00322	0.0150
	(0.00841)	(0.00651)	(0.0309)	(0.0290)	(0.0257)
Muslim * Female	0.0549**	0.0475**	0.0811	0.238*	0.221*
	(0.0234)	(0.0209)	(0.141)	(0.124)	(0.114)
Muslim	0.0903***		0.0328	0.0128	0.00193
	(0.0252)		(0.119)	(0.107)	(0.106)
Mean Dep. Var.	0.80	0.80	0.46	0.64	0.78
Observations	7,731	7,717	1,472	1,472	1,472
Fixed Effect	Village	HH	Village	Village	Village

Notes: The dependent variable for columns (1) and (2) is a dummy that takes value one if the individual uses the latrines for defecation. The samples in the first two columns only include households which had latrines. The dependent variable in each of the columns (3), (4), and (5) is a dummy that takes value one if the respondent reports a toilet as being the topmost, top two or top three priorities, respectively, if money was not a constraint, from a list of assets that the household does not have. The samples in the last three columns only include households without latrines. Columns (1) and (3)-(5) have village fixed effects and control for household level assets and household's main source of income. Column (2) has household level fixed effect. The number of observations in columns (1) and (2) differ with village and household fixed effects as single member households are dropped from the analysis. 'Female' is an indicator variable that takes a value of one for women and zero otherwise. 'Muslim' is an indicator variable that takes a value of one for Muslim households and zero otherwise. The SQUAT states are Haryana, Uttar Pradesh, Madhya Pradesh, Bihar, and Rajasthan. The survey was conducted in December 2013-April 2014. Robust standard errors clustered at village level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C2—No Differential Toilet Preference in Families with Daughters

	Toilet Usage	Toilet Preference		
		Top	Top 2	Top 3
	(1)	(2)	(3)	(4)
Female	0.0856***	0.0420	-0.00554	-0.00222
	(0.0101)	(0.0429)	(0.0402)	(0.0358)
Family has Daughter	-0.00805	-0.0108	-0.0218	-0.0135
	(0.0140)	(0.0464)	(0.0448)	(0.0395)
Female * Family has Daughter	0.0207	0.0171	0.0387	0.0494
	(0.0163)	(0.0592)	(0.0573)	(0.0498)
Mean Dep. Var.	0.80	0.46	0.65	0.78
Observations	7,731	1,472	1,472	1,472
Fixed Effect	Village	Village	Village	Village

Notes: The dependent variable for columns (1) is a dummy that takes value one if the individual uses the latrines for defecation. The samples in the first two columns only include households which had latrines. The dependent variable in each of the columns (2), (3), and (4) is a dummy that takes value one if the respondent reports a toilet as being the topmost, top two or top three priorities, respectively, if money was not a constraint, from a list of assets that the household does not have. The samples in the last three columns only include households without latrines. The SQUAT states are Haryana, Uttar Pradesh, Madhya Pradesh, Bihar, and Rajasthan. All columns have village fixed effects and control for household level assets and household's main source of income. Robust standard errors clustered at village level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C3—Female Autonomy in Female Headed Households

	NFHS Dataset		REDS Dataset	
	Decision	Allowed alone	Village Meeting Attendance	Political Party Activity
	(1)	(2)	(3)	(4)
		Panel A: Overall		
Female Headed HH	0.111***	0.084***	0.036***	0.022***
	(0.006)	(0.005)	(0.014)	(0.007)
		Panel B: Heterogeneity by Religion		
Female Head HH	0.106***	0.083***	0.028**	0.023***
	(0.007)	(0.006)	(0.014)	(0.008)
Female Head HH * Muslim	0.044**	0.011	0.099**	-0.005
	(0.019)	(0.018)	(0.047)	(0.019)
Muslim	0.001	-0.072***	0.005	0.005
	(0.009)	(0.011)	(0.016)	(0.005)
Mean Dep. Var.	0.16	0.59	0.12	0.01
Observations	51,541	70,369	10,342	10,342
Fixed Effect	Village	Village	Village	Village

Notes: The samples in columns (1) and (2) come from the NFHS 2015-16 dataset, while that in columns (3) and (4) come from the REDS 2006 dataset. Column (1) includes all married rural Hindu or Muslim women aged 15-49 while column (2) includes all rural Hindu or Muslim women aged 15-49. Columns (3) and (4) include all (rural) Hindu or Muslim women aged 18 years and above. *Decision* is an indicator variable that takes value one if a respondent can take any of these decision on her own: decisions about health care, decisions about making major household purchases, decisions about visits to family. *Allowed alone* is an indicator variable that takes value one if a respondent is allowed alone to any of these places: market, health facility, outside village. The dependent variable in column (3) is an indicator of whether a woman has attended any of the past four village meetings. The dependent variable in column (4) is an indicator of whether a woman is actively involved with a political party. *Female Headed HH* is an indicator that takes value one if the women respondent belongs to a household headed by a woman. Other controls in all the four columns include age, age squared, years of schooling and its square, religion, caste and village fixed effects. Robust standard errors clustered at village level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C4—Summary Statistics (GP Level)

	Panel A: Demographics
Total population	2672.058 (2044.142)
Primary school within 5 km	.909 (.240)
Middle school within 5 km	.813 (.343)
Secondary school within 5 km	.553 (.459)
Tap water	.186 (.361)
Closed drainage	.056 (.211)
Waste disposal	.092 (.265)
All weather roads	.715 (.415)
Domestic power	.913 (.253)
Irrigation	.871 (.200)
% Women headed hh	.031 (.067)
% APL hh	.668 (.274)
Female reservation 2010	.336 (.472)
Female reservation 2015	.327 (.469)
Muslim share 2010	.141 (.165)
Muslim share 2015	.138 (.165)
	Panel B: Toilet Coverage by Year
Share of HH received toilets in 2013–14	.010 (.070)
Share of HH received toilets in 2014–15	.030 (.110)
Share of HH received toilets in 2015–16	.041 (.129)
Share of HH received toilets in 2016–17	.126 (.255)

Notes: The table reports the summary statistics at the GP level for the full sample of GPs. The variables from "Total Population" to "Irrigation" in Panel A are obtained from the Census 2011 data on village amenities. Gender reservation status in 2010 and 2015 are obtained from UP election data. The remaining variables are obtained using the SBM administrative data. Panel B reports the estimates for the share of eligible households without toilets at the end of the previous year that received toilets in a given year. Standard deviation is reported in parentheses.

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Table C5—Covariates and Pre-treatment Outcomes Balanced (GP Level)

	Female reservation		
	(1)	(2)	(3)
	Panel A: Covariates		
Total population	90.48 (96.66)	108.0 (115.8)	121.9 (146.3)
Primary school within 5 km	0.00521 (0.0175)	0.00883 (0.0210)	0.0237 (0.0269)
Middle school within 5 km	0.0325 (0.0240)	0.0383 (0.0287)	0.0577 (0.0365)
Secondary school within 5 km	0.0196 (0.0321)	0.0149 (0.0385)	0.00365 (0.0491)
Tap water	-0.0236 (0.0243)	-0.0400 (0.0291)	-0.0587 (0.0371)
Closed drainage	0.00348 (0.0134)	0.00671 (0.0160)	0.00675 (0.0202)
Waste disposal	-0.00179 (0.0174)	-0.00470 (0.0207)	-0.00829 (0.0260)
All weather roads	0.0121 (0.0299)	0.0112 (0.0359)	0.0125 (0.0458)
Domestic power	0.0181 (0.0178)	0.0192 (0.0213)	0.0182 (0.0269)
Irrigation	0.00196 (0.0138)	-0.00420 (0.0166)	-0.00627 (0.0210)
% Women headed hh	0.00330 (0.00448)	0.00374 (0.00543)	0.00644 (0.00698)
% APL hh	-0.000357 (0.0191)	0.00621 (0.0229)	0.0159 (0.0293)
Female reservation 2010	0.0119 (0.0355)	0.0170 (0.0429)	-0.00486 (0.0544)
Muslim share 2015	-0.00281 (0.0103)	0.000763 (0.0123)	0.00405 (0.0154)
Δ Muslim share	-0.000123 (0.00322)	0.00169 (0.00384)	0.00109 (0.00480)
	Panel B: Pre-treatment Outcomes		
Share of HH received toilets in 2013–14	-0.00220 (0.00415)	-0.00109 (0.00483)	0.000678 (0.00605)
Share of HH received toilets in 2014–15	-0.00452 (0.00656)	-0.000305 (0.00784)	0.00534 (0.00992)
Share of HH received toilets in 2015–16	0.00980 (0.00926)	0.0105 (0.0111)	0.0117 (0.0141)
Polynomial order	1	1	1
Bandwidth	0.100	0.075	0.050

Notes: The table reports the estimated jump at the threshold of the running variable for female reservation dummy. The variables from "Total Population" to "Irrigation" in Panel A are obtained from the Census 2011 data on village amenities. Gender reservation status in 2010 is obtained from UP election data. The remaining variables are obtained using the SBM administrative data. Panel B reports the estimates for the share of eligible households without toilets at the end of the previous year that received toilets in a given year. Bandwidths used in columns (1), (2), and (3) are 0.100, 0.075, and 0.050 respectively. The polynomial order is 1. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C6—First Stage

	Female reservation		
	(1)	(2)	(3)
Female instrument	0.594***	0.564***	0.528***
	(0.0154)	(0.0178)	(0.0215)
Mean dep. var. in control GPs	0.035	0.041	0.045
Observations	2,470,191	1,962,725	1,457,226
Number of GPs	9,179	7,234	5,277
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The polynomial order is 1. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C7—Muslim Share and Gender Gap in Toilet Preference for Hindus and Muslims

	Toilet Usage		Toilet Preference		
			Top	Top 2	Top 3
	(1)	(2)	(3)	(4)	(5)
Female	0.0832***	0.0825***	0.0397	-0.0117	0.00197
	(0.0109)	(0.00945)	(0.0339)	(0.0322)	(0.0273)
Muslim	0.125		0.108	0.0413	0.0129
	(0.0842)		(0.192)	(0.179)	(0.171)
Female * Muslim share	0.121	0.170**	0.184	0.332	0.253
	(0.0790)	(0.0705)	(0.288)	(0.264)	(0.213)
Muslim * Female * Muslim share	-0.0387	-0.109	0.788	-0.122	-0.0235
	(0.151)	(0.127)	(0.600)	(0.604)	(0.376)
Mean Dep. Var.	0.80	0.80	0.46	0.65	0.78
Observations	7,731	7,717	1,472	1,472	1,472
Fixed Effect	Village	HH	Village	Village	Village

Notes: The dependent variable for columns (1) and (2) is a dummy that takes value one if the individual uses the latrines for defecation. The samples in the first two columns only include households which had latrines. The dependent variable in each of the columns (3), (4), and (5) is a dummy that takes value one if the respondent reports a toilet as being the topmost, top two or top three priorities, respectively, if money was not a constraint, from a list of assets that the household does not have. The samples in the last three columns only include households without latrines. All two-way interactions are included in all specifications. Columns (1) and (3)-(5) have village fixed effects and control for household level assets and household's main source of income. Column (2) has household level fixed effect. The number of observations in columns (1) and (2) differ with village and household fixed effects as single member households are dropped from the analysis. 'Female' is an indicator variable that takes a value of one for women and zero otherwise. 'Muslim share' is the proportion of Muslims in a village estimated from the SQUAT survey. 'Muslim' is an indicator variable that takes a value of one for Muslim households and zero otherwise. The SQUAT states are Haryana, Uttar Pradesh, Madhya Pradesh, Bihar, and Rajasthan. The survey was conducted in December 2013-April 2014. Robust standard errors clustered at village level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C8—First Stage: Heterogeneity

	Panel A: Female reservation		
	(1)	(2)	(3)
Female instrument	0.629***	0.593***	0.549***
	(0.0201)	(0.0234)	(0.0285)
Female instrument*Muslim share	-0.274***	-0.234*	-0.168
	(0.104)	(0.121)	(0.150)
Muslim Share	-0.0247	-0.0148	-0.00837
	(0.0384)	(0.0466)	(0.0592)
	Panel B: Female reservation*Muslim Share		
	(1)	(2)	(3)
Female instrument	0.00617	0.00389	0.00116
	(0.00483)	(0.00550)	(0.00655)
Female instrument*Muslim share	0.500***	0.494***	0.491***
	(0.0547)	(0.0626)	(0.0748)
Muslim Share	0.0378**	0.0451**	0.0546**
	(0.0163)	(0.0200)	(0.0256)
Observations	2,470,191	1,962,725	1,457,226
Number of GPs	9,179	7,234	5,277
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The polynomial order is 1. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C9—Heterogeneous Impact of Female Reservation on Toilet Allocation: Quadratic

	HH Received Toilet 2016–17		
	(1)	(2)	(3)
Female reservation	-0.0378	-0.0466	-0.0505
	(0.0325)	(0.0381)	(0.0462)
Female reservation*Muslim share	0.577***	0.648***	0.541*
	(0.216)	(0.245)	(0.292)
Muslim share	-0.137**	-0.138*	-0.0713
	(0.0663)	(0.0755)	(0.0886)
Mean dep. var. in control GPs	0.107	0.107	0.100
Observations	2,470,191	1,962,725	1,457,226
Number of GPs	9,179	7,234	5,277
Polynomial order	2	2	2
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if the household received a toilet in 2016–17, and zero otherwise. The sample includes eligible households, i.e., those that did not have toilet at the end of 2015–16 and were eligible to receive toilet under the SBM scheme. The polynomial order is 2. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table C10—Correlation between Covariates and Muslim Share

	Coefficient	SE	Observations	R-squared
(1)	(2)	(3)	(4)	(5)
Total population	569.4***	(90.61)	9,203	0.004
Primary school within 5 km	0.0613***	(0.0165)	9,084	0.002
Middle school within 5 km	-0.0174	(0.0232)	9,084	0.000
Secondary school within 5 km	-0.0266	(0.0311)	9,084	0.000
Tap water	0.000964	(0.0239)	9,026	0.000
Closed drainage	-0.00391	(0.0137)	9,032	0.000
Waste disposal	-0.0125	(0.0169)	9,016	0.000
All weather roads	0.0872***	(0.0288)	9,026	0.001
Domestic power	-0.0519***	(0.0179)	9,067	0.001
Irrigation	0.0410***	(0.0139)	9,045	0.001
Women headed hh share	-0.00300	(0.00443)	9,117	0.000
APL hh share	0.0142	(0.0184)	9,203	0.000

Notes: The table reports the estimates after regressing the variables mentioned in column (1) on the proportion of Muslim households in a Gram Panchayat. We include the Gram Panchayats within a bandwidth of 10% around the threshold. Each row is a separate regression and the number of Gram Panchayats included in estimation are given in column (4). The variables from "Total Population" to "Irrigation" in Panel A are obtained from the Census 2011 data on village amenities. The remaining variables are obtained using the SBM administrative data. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C11—Covariates and Pre-treatment Outcomes Balanced (GP Level): By Muslim Share

	Female reservation*Muslim share		
	(1)	(2)	(3)
	Panel A: Covariates		
Total population	502.1 (668.1)	350.6 (765.9)	-91.69 (928.3)
Primary school within 5 km	-0.313** (0.122)	-0.342** (0.146)	-0.414** (0.188)
Middle school within 5 km	-0.263 (0.186)	-0.348 (0.218)	-0.397 (0.274)
Secondary school within 5 km	-0.0742 (0.246)	-0.238 (0.292)	-0.519 (0.377)
Tap water	-0.0772 (0.172)	-0.0868 (0.200)	-0.125 (0.249)
Closed drainage	0.0971 (0.112)	0.0636 (0.129)	0.0159 (0.155)
Waste disposal	0.0671 (0.132)	0.135 (0.148)	0.260 (0.172)
All weather roads	0.0411 (0.211)	0.0550 (0.247)	-0.0220 (0.310)
Domestic power	-0.177 (0.143)	-0.246 (0.166)	-0.362* (0.206)
Irrigation	0.139 (0.105)	0.0836 (0.121)	0.0646 (0.146)
Women headed hh share	0.0557* (0.0303)	0.0572 (0.0363)	0.0611 (0.0456)
APL hh share	-0.215 (0.155)	-0.215 (0.183)	-0.308 (0.229)
Female reservation 2010	0.226 (0.265)	0.421 (0.314)	0.522 (0.394)
Δ Muslim share	-0.0788 (0.0509)	-0.0906 (0.0604)	-0.110 (0.0781)
	Panel B: Pre-treatment Outcomes		
Share of HH received toilets in 2013–14	-0.00810 (0.0237)	-0.0123 (0.0264)	0.00407 (0.0290)
Share of HH received toilets in 2014–15	-0.0361 (0.0409)	-0.0376 (0.0449)	-0.0568 (0.0509)
Share of HH received toilets in 2015–16	-0.0428 (0.0941)	-0.0510 (0.115)	-0.113 (0.150)
Polynomial order	1	1	1
Bandwidth	0.100	0.075	0.050

Notes: The table reports the estimated jump at the threshold of the running variable for the interaction of female reservation with Muslim share. The variables from "Total Population" to "Irrigation" in Panel A are obtained from the Census 2011 data on village amenities. Gender reservation status in 2010 is obtained from UP election data. The remaining variables are obtained using the SBM administrative data. Panel B reports the estimates for the share of eligible households without toilets at the end of the previous year that received toilets in a given year. Bandwidths used in columns (1), (2), and (3) are 0.100, 0.075, and 0.050 respectively. The polynomial order is 1. Standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C12—Heterogeneous Impact of Female Reservation on Toilet Allocation: Robustness with Controls

	HH Received Toilet 2016–17		
	(1)	(2)	(3)
Female reservation	0.272	0.295	0.305
	(0.323)	(0.412)	(0.572)
Female reservation*Muslim share	0.343**	0.472***	0.635***
	(0.145)	(0.175)	(0.229)
Muslim share	-0.0336	-0.0859	-0.130*
	(0.0491)	(0.0574)	(0.0738)
Mean dep. var. in control GPs	0.105	0.104	0.0986
Observations	2,370,649	1,884,179	1,399,694
Number of GPs	8,768	6,911	5,049
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if the household received a toilet in 2016–17, and zero otherwise. The sample includes eligible households, i.e., those that did not have toilet at the end of 2015–16 and were eligible to receive toilet under the SBM scheme. We control for total population of the GP, proportion of villages in the GP with primary, middle, and secondary school within 5 km, access to tap water, closed drainage, waste disposal, all weather roads, domestic power, irrigation, proportion of APL households and female headed households, and their interactions with the female reservation indicator. The polynomial order is 1. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table C13—No Heterogeneous Impact of Female Reservation on Electing Muslim Leader

	Sarpanch is Muslim		
	(1)	(2)	(3)
Female reservation	-0.0207	-0.0261	-0.0203
	(0.0164)	(0.0194)	(0.0249)
Female reservation * Muslim share	0.0661	0.0696	0.0740
	(0.203)	(0.238)	(0.299)
Muslim share	1.246***	1.263***	1.271***
	(0.0816)	(0.0935)	(0.113)
Mean dep. var. in control GPs	0.107	0.107	0.100
Observations	9,179	7,234	5,277
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if the elected leader or Sarpanch of the GP is Muslim, and is zero otherwise. The sample includes GPs within the bandwidth of 0.1. The polynomial order is 1. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C14—Variation in Muslim Share due to Delimitation and Female Reservation Effect

	HH Received Toilet 2016–17		
	(1)	(2)	(3)
Female reservation	-0.0314	-0.0390	-0.0486
	(0.0239)	(0.0285)	(0.0358)
Female reservation * Muslim share 2010	0.338**	0.431**	0.538**
	(0.172)	(0.203)	(0.250)
Female reservation * Δ Muslim share	0.727**	0.789**	0.911*
	(0.323)	(0.398)	(0.500)
Muslim share 2010	-0.0210	-0.0559	-0.0860
	(0.0565)	(0.0643)	(0.0774)
Δ Muslim share	-0.323**	-0.299**	-0.265**
	(0.131)	(0.133)	(0.129)
Mean dep. var. in control GPs	0.107	0.107	0.100
Observations	1,854,500	1,468,930	1,088,135
Number of GPs	7,488	5,892	4,322
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if a household received toilet in 2016–17, and zero otherwise. The sample includes eligible households, i.e., those that did not have toilet at the end of 2015–16 and were eligible to receive toilet under the SBM scheme. The polynomial order is 1. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table C15—Gender Quota Effect in NREGS Implementation: No Heterogeneity

	Expenditure/Capita 2016–17					
	(1)	(2)	(3)	(4)	(5)	(6)
Female reservation	0.00648	0.00648	0.00913	-0.00966	-0.00935	-0.00556
	(0.0240)	(0.0289)	(0.0372)	(0.0319)	(0.0383)	(0.0493)
Female reservation*Muslim share				0.139	0.141	0.132
				(0.155)	(0.176)	(0.214)
Muslim share				-0.131**	-0.173**	-0.212**
				(0.0597)	(0.0681)	(0.0848)
Mean dep. var. in control GPs	0.297	0.294	0.289	0.297	0.294	0.289
Observations	8,648	6,822	4,978	8,648	6,822	4,978
Polynomial order	1	1	1	1	1	1
Bandwidth	0.100	.075	0.050	0.100	.075	0.050

Notes: The dependent variables is the annual expenditure under NREGS per capita during 2016–17 in the GP. The expenditure reported is in thousands of ₹. Columns 1–3 look at the overall effect while Columns 3–6 look at heterogeneity with respect to muslim population share in 2015. Coefficients for Female reservation reports the RDD estimates for three different bandwidths. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C16—Female vs Male Headed Households

	(1) Poor	(2) Education of members	(3) Household members	(4) Age of members	(5) Child proportion	(6) Old proportion
			Panel A: Overall			
Female Headed HH	0.072***	-0.796***	-1.174***	1.724***	-0.011***	0.054***
	(0.002)	(0.014)	(0.010)	(0.079)	(0.001)	(0.001)
			Panel B: Heterogeneity by Religion			
Female Headed HH	0.074***	-0.816***	-1.160***	1.869***	-0.014***	0.057***
	(0.002)	(0.015)	(0.011)	(0.083)	(0.001)	(0.002)
Muslim*Female Headed HH	-0.019***	0.242***	-0.161***	-1.316***	0.023***	-0.024***
	(0.006)	(0.041)	(0.036)	(0.232)	(0.004)	(0.004)
Muslim	-0.008*	-0.904***	0.493***	-2.657***	0.045***	-0.020***
	(0.004)	(0.029)	(0.022)	(0.113)	(0.002)	(0.002)
Observations	425,563	425,532	425,563	425,549	425,563	425,563
Fixed Effect	Village	Village	Village	Village	Village	Village

Notes: The sample includes rural data from the NFHS 2015-16 dataset for all states. *Female Headed HH* is an indicator variable that takes value one if a household is headed by a woman, and is zero otherwise. *Poor* is an indicator variable that takes value one if a household belongs to the poorest two quintiles of the wealth index distribution. *Education of members* is average years of education of all household members. *Age of members* is average years of age of all household members. *Child proportion* and *Old proportion* refer to proportion of children (age 0-14) and old (age 60 and above) among all household members, respectively. *Muslim* is an indicator variable that takes value one if a household's head belongs to Muslim religion, else zero. Village fixed effects are included as a control in all specifications. Robust standard errors clustered at village level are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C17—No Female Reservation Effect for SC/ST Households

	Dep. Var.: HH Received Toilet in 2016–17		
	(1)	(2)	(3)
Female reservation	0.0233	0.0356	0.0542
	(0.0226)	(0.0272)	(0.0342)
Female reservation * SC/ST HH	0.00427	0.00261	0.00155
	(0.0266)	(0.0313)	(0.0386)
SC/ST HH	-0.0413***	-0.0417***	-0.0437***
	(0.0113)	(0.0126)	(0.0147)
Mean dep. var. in control GPs	0.120	0.120	0.112
Observations	1,602,562	1,264,590	939,209
Number of GPs	8,887	7,005	5,113
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is a dummy that takes value one if the household received a toilet in 2016–17, and zero otherwise. The sample includes Above Poverty Line (APL) eligible households,i.e., those that did not have toilet at the end of 2015–16 and were eligible to receive toilet under the SBM scheme. The polynomial order is 1. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table C18—Impact of Female Reservation on Female Headed Households: Panel Estimation

	Dep. Var.: HH Received Toilet					
	Hindu Households			**Muslim Households**		
	(1)	(2)	(3)	(4)	(5)	(6)
Post * Female reservation	0.0172	0.0259	0.0393	0.0318	0.0564	0.111*
	(0.0187)	(0.0221)	(0.0276)	(0.0403)	(0.0480)	(0.0594)
Post * Female reservation * Female Headed Household	0.0569	0.0610	0.0365	0.291***	0.333***	0.314**
	(0.0468)	(0.0561)	(0.0685)	(0.104)	(0.118)	(0.138)
Post * Female Headed Household	-0.0303	-0.0300	-0.0157	-0.0758***	-0.0709***	-0.0436
	(0.0188)	(0.0213)	(0.0236)	(0.0249)	(0.0257)	(0.0281)
Post	0.107***	0.104***	0.100***	0.122***	0.107***	0.0928***
	(0.00802)	(0.00905)	(0.0107)	(0.0161)	(0.0182)	(0.0216)
Household Fixed Effects	YES	YES	YES	YES	YES	YES
Mean dep. var. in control GPs	0.09	0.10	0.09	0.12	0.12	0.12
Observations	2,209,092	1,736,072	1,289,674	309,536	243,158	175,024
Number of GPs	7,202	5,657	4,152	6,306	4,943	3,630
Polynomial order	1	1	1	1	1	1
Bandwidth	0.100	.075	0.050	0.100	0.075	0.050

Notes: The dependent variable is a dummy that takes value one if a household received toilet, and zero otherwise. The sample includes above poverty line (APL) eligible households, i.e., those that did not have toilet at the end of 2014-15 and were eligible to receive toilet under the SBM scheme. The first and last three columns are Hindu and Muslim households, respectively. Female Headed Household is a dummy that takes value one if the household head is female and is zero otherwise. The polynomial order is 1. Standard errors are clustered at GP level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C19—Female Reservation Effect in NREGS Implementation: No Heterogeneity (Robustness)

	Expenditure/Capita 2016–17		
	(1)	(2)	(3)
Female reservation	0.00443	0.00395	0.0161
	(0.0358)	(0.0431)	(0.0553)
Female reservation*Muslim share 2010	0.131	0.127	0.0627
	(0.180)	(0.202)	(0.242)
Female reservation*Δ Muslim share	-0.165	-0.236	0.0404
	(0.483)	(0.585)	(0.699)
Muslim share 2010	-0.157**	-0.207***	-0.241**
	(0.0679)	(0.0777)	(0.0979)
Δ Muslim share	0.211	0.195	0.0742
	(0.204)	(0.227)	(0.261)
Mean dep. var. in control GPs	0.298	0.294	0.288
Observations	7,097	5,600	4,108
Polynomial order	1	1	1
Bandwidth	0.100	.075	0.050

Notes: The dependent variable is the annual expenditure under NREGS per capita during 2016–17 at the GP level. The expenditure reported is in thousands of ₹. RDD estimates for three different bandwidths. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table C20—Muslim vs Hindu Sarpanches in Close Election GPs

	HH Received Toilet in 2016–2017		
	(1)	(2)	(3)
Female reservation	-0.0143	-0.0256	-0.0298
	(0.0242)	(0.0274)	(0.0339)
Muslim Sarpanch	-0.0204	-0.0261	-0.0107
	(0.0216)	(0.0247)	(0.0302)
Female reservation * Muslim Sarpanch	0.00658	0.0191	0.0219
	(0.0349)	(0.0399)	(0.0492)
Mean dep. var. in control GPs	0.099	0.098	0.108
Observations	943,640	777,713	569,103
Number of GPs	3,263	2,666	1,941
Bandwidth	0.100	0.075	0.050
Polynomial order	1	1	1

Notes: The sample of GPs included had a close election between a Hindu and a Muslim candidate. The absolute value of the margin of victory of the Muslim candidate is 0.1 in column (1), 0.075 in column (2) and 0.05 in column (3). The sample only includes eligible households that did not have toilets at the end of 2015–2016. The polynomial order is 1. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C21—Covariates and Pre-treatment Outcomes Balanced (Close Election; GP Level)

	Muslim Sarpanch		
	(1)	(2)	(3)
	Panel A: Covariates		
Total population	-146.3 (185.3)	-135.5 (213.9)	-66.29 (268.5)
Primary school within 5 km	0.00770 (0.0152)	0.0131 (0.0172)	0.0155 (0.0205)
Middle school within 5 km	-0.0109 (0.0246)	-0.00202 (0.0280)	0.00436 (0.0338)
Secondary school within 5 km	-0.00932 (0.0323)	-0.0210 (0.0366)	-0.00227 (0.0441)
Tap water	0.0127 (0.0261)	0.0150 (0.0297)	0.0144 (0.0356)
Closed drainage	-0.00405 (0.0166)	-0.00210 (0.0188)	-0.00402 (0.0224)
Waste disposal	-0.0215 (0.0198)	-0.0272 (0.0227)	-0.0459* (0.0279)
All weather roads	-0.0205 (0.0276)	-0.0121 (0.0311)	0.0118 (0.0372)
Domestic power	-0.00683 (0.0190)	-0.0135 (0.0217)	-0.0153 (0.0264)
Irrigation	-0.00333 (0.0140)	-0.00383 (0.0156)	-0.00188 (0.0182)
% Women headed hh	0.000520 (0.00322)	0.00235 (0.00367)	0.000339 (0.00454)
% APL hh	-0.0174 (0.0197)	-0.0138 (0.0224)	-0.0156 (0.0268)
Female reservation 2010	0.0341 (0.0362)	0.0420 (0.0409)	0.0649 (0.0493)
Female reservation 2015	-0.0360 (0.0335)	-0.0215 (0.0380)	0.0225 (0.0456)
Muslim share 2015	0.0226* (0.0117)	0.0230* (0.0132)	0.0310* (0.0159)
Δ Muslim share	-0.00178 (0.00517)	0.000727 (0.00590)	0.00469 (0.00709)
	Panel B: Pre-treatment Outcomes		
Share of HH received toilets in 2013–14	0.00252 (0.00463)	0.00296 (0.00517)	0.000438 (0.00619)
Share of HH received toilets in 2014–15	-0.00135 (0.00820)	-0.00395 (0.00937)	-0.0101 (0.0116)
Share of HH received toilets in 2015–16	0.00426 (0.00907)	0.00163 (0.0103)	0.00499 (0.0125)
Polynomial order	1	1	1
Bandwidth	0.100	0.075	0.050

Notes: The table reports the estimated jump at the threshold of Muslim Sarpanch victory. The variables from "Total Population" to "Irrigation" in Panel A are obtained from the Census 2011 data on village amenities. Gender reservation status in 2010 and 2015 are obtained from UP election data. The remaining variables are obtained using the SBM administrative data. Panel B reports the estimates for the share of eligible households without toilets at the end of the previous year that received toilets in a given year. Bandwidths used in columns (1), (2), and (3) are 0.100, 0.075, and 0.050 respectively. The polynomial order is 1. Standard errors are clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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Table C22—Muslim vs Hindu Sarpanches in Close Election GPs: Robustness

	HH Received Toilet in 2016–2017		
	(1)	(2)	(3)
Female Reservation	-0.00780	-0.0166	-0.0171
	(0.0258)	(0.0294)	(0.0365)
Muslim Sarpanch	-0.0194	-0.0249	-0.00972
	(0.0216)	(0.0247)	(0.0302)
Female Reservation*Muslim Sarpanch	0.00389	0.0162	0.0205
	(0.0352)	(0.0402)	(0.0497)
Mean dep. var. in control GPs	0.099	0.098	0.108
Observations	943,224	777,713	569,103
Number of GPs	3,262	2,666	1,941
Bandwidth	0.100	0.075	0.050
Polynomial order	1	1	1

Notes: The sample only includes GPs that had a close election between a Hindu and a Muslim candidate. The sample is restricted to households which did not have toilets at the end of 2015–16. The polynomial order is 1. All regressions additionally control for the reservation assignment variable and its interaction with female reservation. The bandwidth is 0.1. Standard errors clustered at Gram Panchayat level and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

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